

Astryx

User Guide

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Chapter 1

Introduction

Plan better. Image more. Waste less clear sky.

Astryx helps you decide what to image, when to image it, and how long to image it. It searches a database of over 14,000 deep-sky objects, works out when each one is well placed from your location, builds an imaging plan for the night, keeps a log of your imaging projects, and analyzes the ASIAir and PHD2 logs from your sessions.

Everything starts at astryx.tools. From there you can open the browser version and start using it right away, with nothing to install, or go to the **Downloads** page for the desktop app, available for Windows and for Ubuntu-based Linux systems.

> **Quick Start**
> Add your observing location and your equipment, and Astryx is ready to use. You don't need to read anything else first. Getting Started has the details.
> Open the system menu (≡), choose **Start Tutorial**, and follow along. It walks you through both steps and takes about ten minutes.

Getting Help as You Go

Every tutorial is listed under **Tutorials** in the system menu. Each one walks you through a single view in a few minutes, and when it finishes it offers to start the next. If you take them in order, you'll have seen everything Astryx does. If you leave a tutorial partway through, **Resume Tutorial** in the system menu picks up where you stopped. It first asks you to open the tutorial's view, or the menu or dialog you were in, so you can resume from anywhere in Astryx.

Hold the pointer over a control and a tooltip explains what it does.

This guide is for when you want more than a tooltip: what a view is for, how to read its results, and what to do next. Recipes walks through common jobs from start to finish, and Something Doesn't Look Right explains the most common surprises. The Glossary defines the terms.

What Do You Want to Do?

Start from your question and go to the view that answers it.

I want to...	Go to
Find targets by name, type, size, or catalog	Target Selection
Find targets that are well placed in a given month	Target Selection, Visibility filter
Keep a shortlist of targets I want to image	To Do List
See when during the year a target is well placed	Yearly Observability
See a target's altitude through one night, with the Moon and weather	Daily Visibility
See how a target fits my camera's field of view	Viewfinder
Find out what is best to image on a given night	Target Optimizer
Plan the order and timing of several targets in one night	Sequence Planner
Keep track of imaging projects and sessions	Imaging Log
Find out what went wrong, or right, in a session	Log Analysis
Check cloud cover and light pollution	Utilities
Save my data, or move it to another computer	Backup and Restore

Who Astryx Is For

Astryx was built by an imager who uses one-shot-color cameras and an ASIAir, and it works best for that kind of setup. If you image in mono, or control your rig with NINA, most of Astryx works just as well for you, but you may find gaps, particularly in the session details that the Imaging Log records, and in Log Analysis, which reads ASIAir and PHD2 logs.

If you'd like Astryx to work better for mono or NINA imaging, please describe what you need, or what you can contribute, in an issue at github.com/sparsileg/astryx/issues.

Your Data Stays With You

Whether you use the browser version or the desktop app, everything you enter is stored on your own computer and nowhere else. No account is needed, and your targets, plans, and logs are never sent to a remote server.

That also means nobody else can recover your data if it's lost. Use **Backup** in the system menu regularly, or turn on automatic backups in **Settings**.

> **Why this matters**
> In the web version, your data lives in the browser, and a browser can clear it, for example when you clear its cache. Safari may delete it after a week without a visit, so use a different browser for Astryx. The desktop app keeps your data in its own database, which clearing a browser doesn't touch. Either way, keep backups: a backup is the only copy that survives a failed disk or a lost laptop.

Astryx works without an internet connection, which makes it useful in the field. A few things need a connection when you use them: the weather forecast in Daily Visibility, the sky images in the

Viewfinder, the cloud and light-pollution websites that Utilities opens, and the Wikipedia links in a target's details. These send only your location's coordinates, or the target's position or name. Sky images you've already viewed are kept for about two weeks, so they still work offline. This guide opens from astryx.tools, so it needs a connection too; save a copy from there to read it offline.

Moving Around

The sidebar on the left lists the views. Above them are the color theme, the location you're observing from, and the **Current Target** button. Below them is the list of your **Pinned Targets**. Drag the sidebar's right edge to widen or narrow it, and double-click the edge to return to the default width. The arrow at its top collapses it to give the views more room.

The Current Target

Several views work on one target at a time: the Current Target, whose name is on the button at the top of the sidebar. Daily Visibility, Yearly Observability, and the Viewfinder all show it, so you pick a target once and then move between those views. To change the Current Target:

- In Target Selection, click a target in the results list.
- Click a target in the To Do List.
- Click a target's name under **Pinned Targets** in the sidebar.
- Click **Daily Visibility** on a Target Optimizer result card.

The first three also open the target's **Target Details** window, with its designations, size, coordinates, and observability, and the **Pin** and **Add to To Do List** buttons (see Finding Targets). Click the **Current Target** button to open it again.

In Daily Visibility, Yearly Observability, and the Viewfinder, clicking a pinned target doesn't open Target Details; the view switches to it at once. Click between your Pinned Targets to compare them.

The system menu (≡) at the top of the sidebar holds **Settings**, Admin Tools (locations, equipment, and target database maintenance), Backup and Restore, the tutorials, and **Help**, which opens this guide from astryx.tools.

Chapter 2

Getting Started

> **At a glance**
> Astryx needs two things from you: where you observe from, and what you image with. Add a location and your equipment, and every view is ready to use. **Start Tutorial** in the system menu (≡) walks you through both in about ten minutes.

Add Your Location

Almost everything Astryx calculates, including dusk and dawn, rise and set times, and how high a target climbs, depends on where you are. Add each place you image from: your backyard, a dark site, a star party field.

1. Open the system menu (≡) and choose **Admin Tools**, then **Manage Observer Locations**.
2. Fill in the **Add New Location** form (the fields are described below).
3. Click **Save Location**.
4. Choose the new location from the location list at the top of the sidebar.

The first time you choose a location, Astryx calculates the best month to image each target from there. A progress message shows how far it has got. This happens once per location, so a site you add later gets the same treatment the first time you choose it.

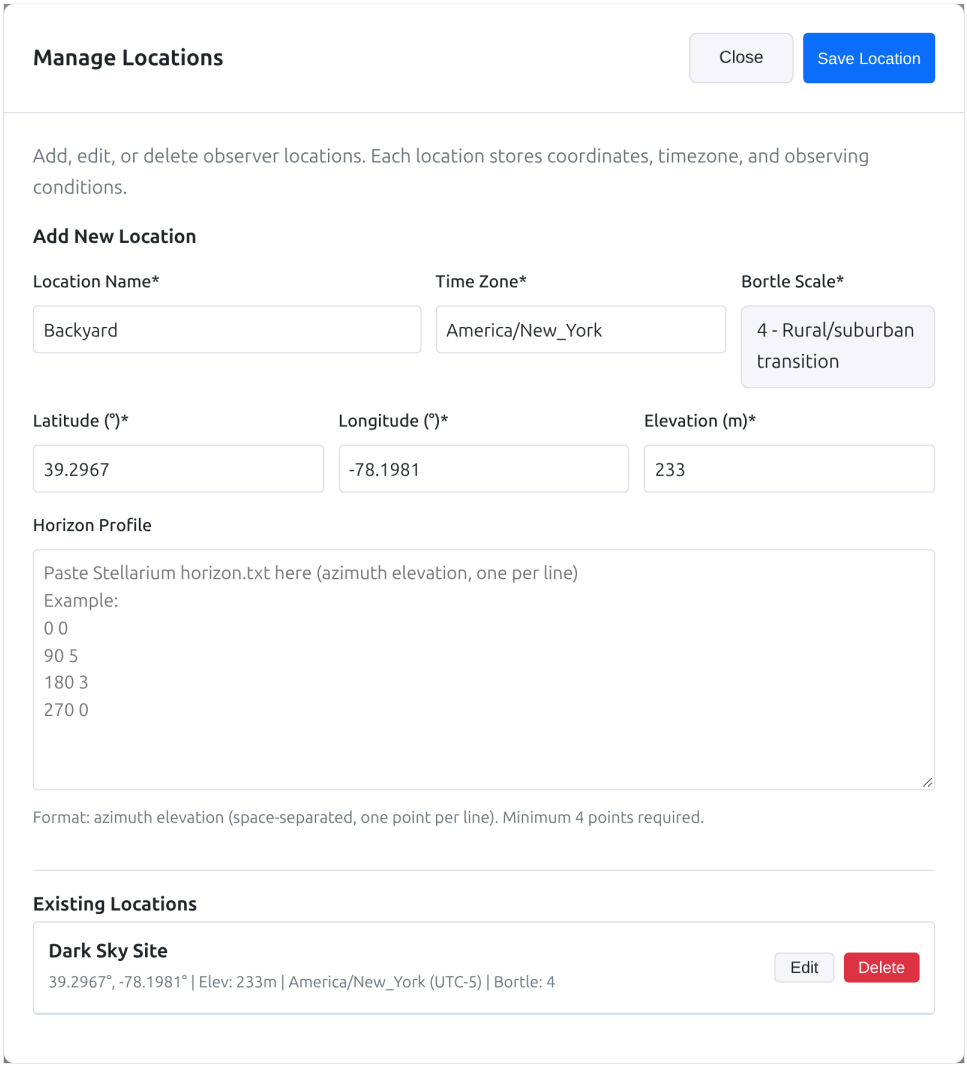


Figure 1: The Manage Locations dialog with a location filled in, showing the Existing Locations list below.

The location fields

Field	What to enter
Location Name	Any name you'll recognize, such as "Backyard" or "Cherry Springs".
Time Zone	The location's time zone by name, such as America/New_York or Europe/Madrid. It starts out as your computer's zone; start typing to pick another from the list.
Bortle Scale	How light-polluted the sky is, from 1 (excellent dark sky) to 9 (inner city). It's kept with the location for your reference.
Latitude	Decimal degrees, positive north of the equator and negative south.
Longitude	Decimal degrees, positive east of Greenwich and negative west. Anywhere in the Americas is negative.
Elevation	Meters above sea level.
Horizon Profile	Optional. The height of the trees, buildings, and hills around you. See below.

Why this matters
Astryx uses the time zone's own daylight saving rules, not your computer's. A location in Arizona never shifts its clocks, even if your laptop is set to Denver time, and a location in Europe changes its clocks on the European dates. Pick the zone by name and daylight saving takes care of itself.

Astryx Tip
To find your latitude and longitude, right-click your observing spot in Google Maps: the first item in the menu is the coordinates, in decimal degrees, ready to copy. If you're not sure of your Bortle class, the **Light Pollution** link in Utilities → **Sky Conditions** shows it.

Horizon profile

A horizon profile tells Astryx where your view of the sky is blocked. Without one, Astryx assumes a flat horizon all the way around, and a target is high enough to image as soon as it clears your minimum altitude.

Enter the profile as one point per line: the azimuth (the compass direction, 0 for north, 90 for east, 180 for south, 270 for west) and the height of the obstruction in that direction, in degrees, separated by a space. You need at least four points. Astryx draws a straight line between neighboring points, so add more points where the skyline changes quickly.

```
0 12
45 25
90 18
135 8
180 5
225 10
270 30
315 22
```

This is the same format as a Stellarium `horizon.txt` file, so if you've already made a Stellarium landscape for your site, you can paste its points in directly. Leave the box empty for a flat horizon. Astryx will automatically create a 4-point horizon that has no elevation.

A target has to clear both the horizon profile and your minimum altitude. The horizon is used by Daily Visibility (which can switch it off), the Sequence Planner (as an option), and the Target Optimizer. Best months and Yearly Observability use only the minimum altitude.

Astryx Tip
You don't need special equipment to measure a horizon. On an Android phone, the AngleCam app works well: stand where your telescope sits, point the camera along the skyline, and it shows the compass direction and height of whatever is in the crosshairs. Note the height of the skyline whenever the elevation begins to change. A compass app and an inclinometer app do the same job on any phone. Ten minutes of work gives you a profile good enough to stop Astryx from scheduling a target that's behind your neighbor's oak tree.

Changing or removing a location

Click **Edit** next to a location in the Existing Locations list to load it into the form, change what you need, and click **Save Location**. A location's name can't be changed once it's saved; to use a different name, add a new location and delete the old one. A new location can't reuse a saved location's name, even with different capitals or spaces. Click **Delete** to remove it. A location that's recorded in an Imaging Log session can't be deleted.

Add Your Equipment

Astryx uses your telescope and camera to show how a target fits your field of view. Open the system menu (≡) and choose **Admin Tools**, then **Manage Equipment**. The dialog has three tabs.

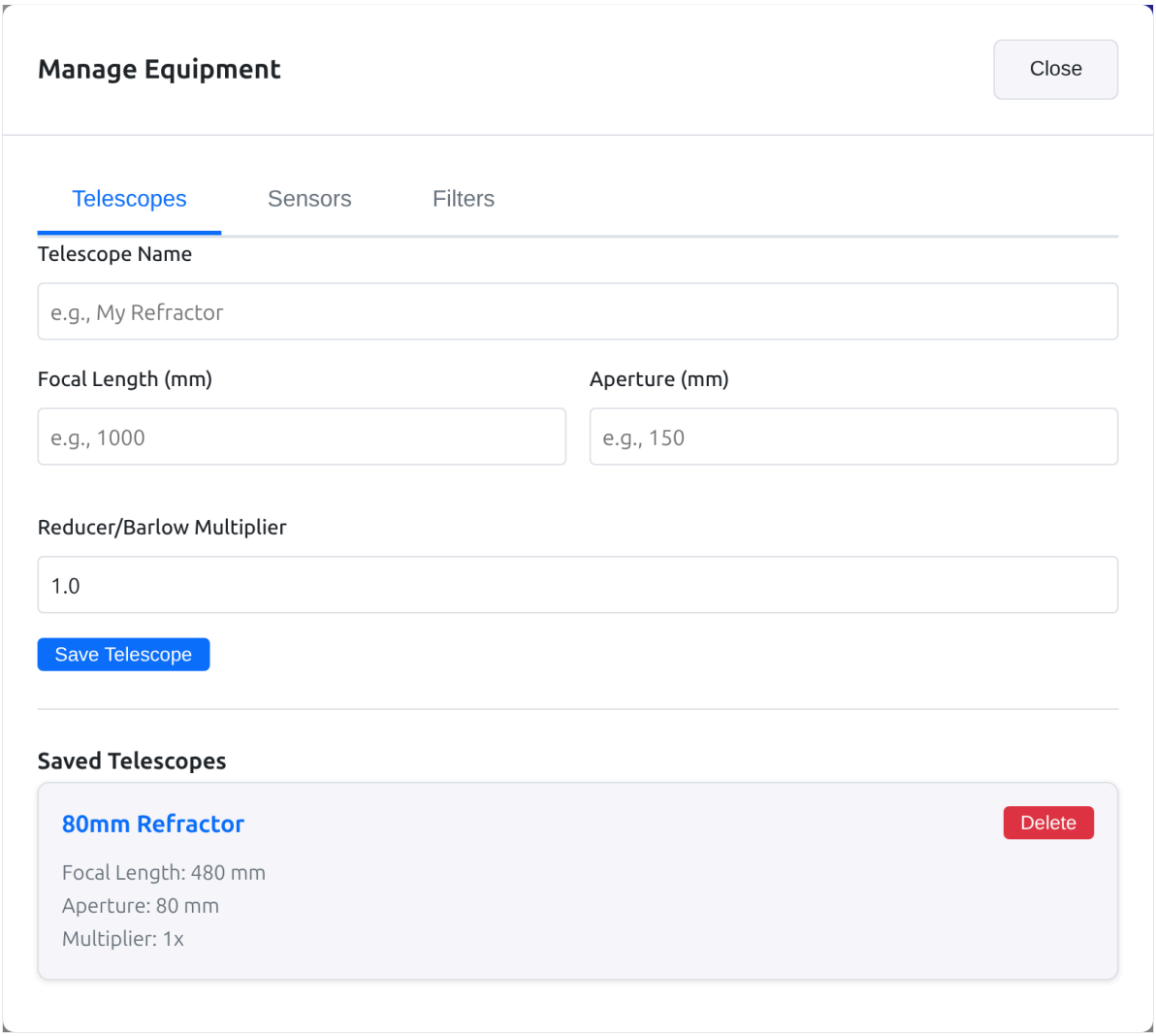


Figure 2: The Manage Equipment dialog, Telescopes tab, with one saved telescope.

Telescopes

Field	What to enter
Telescope Name	A name you'll recognize.

Field	What to enter
Focal Length	The telescope's native focal length, in millimeters.
Aperture	In millimeters.
Reducer/Barlow Multiplier	1.0 with nothing in the light path. 0.8 for a 0.8× reducer, 2.0 for a 2× Barlow.

Click **Save Telescope**.

> **Astryx Tip**
> If you use the same telescope with and without a reducer, save it twice, once for each setup: for example, naming it "Esprit 100" at 1.0 and "Esprit 100 + 0.79×" at 0.79. Then you can pick the configuration you'll actually use.

Sensors

Enter a name, the sensor's resolution in pixels (X across, Y down), and its pixel size in microns. The camera maker's specification sheet has all four numbers. Click **Save Sensor**.

Filters

Filters only need a name, such as L-eXtreme, Ha, or OIII. You select them when you record a session in the Imaging Log, and every session needs one, so add at least one before your first session.

Changing or removing equipment

Equipment can't be edited in place. To change a telescope or sensor, save it again under the same name with the new values. Astryx asks before replacing the old entry; capitals and spaces don't make it a different name.

Imaging Log sessions refer to equipment by name. If a session used the telescope or sensor, the session shows the new values after you replace it, and Astryx says so when it asks. To keep the old values for past sessions, save the changed setup under a new name instead.

A filter can't be added twice under the same name. Click **Delete** to remove an item. Equipment mentioned in an Imaging Log session can't be deleted.

> **Note**
> Only the Viewfinder, the dust mote calculator in Utilities, and the Imaging Log use your equipment. Every other view needs nothing but a location.

Settings

Everything in Settings has a sensible default, so you can leave it alone until you have a reason to change it. Open the system menu (≡), choose **Settings**, make your changes, and click **Save Settings**.

Setting	Default	What it does
DST Mode	Automatic	Automatic follows each location's time zone. **Always Active** and **Never Active** override it for every location; you shouldn't need them.
Minimum Altitude	35°	The lowest a target can be and still be high enough to image. It's the starting value in Daily Visibility, Yearly Observability, and the Sequence Planner, and the value the Target Optimizer, To Do List, and Visibility filter use. Changing it means recalculating Best Months; Astryx offers to.
Date Format	Your computer's region	How dates are shown and typed: **mm/dd/yyyy**, **dd/mm/yyyy**, or **yyyy-mm-dd**. Astryx always stores dates as yyyy-mm-dd, so changing it changes only what you see.
Min Target Size	4.0′	The default minimum size, in arcminutes, for the Target Selection filters. Default value can be changed in the search view.
Limiting Magnitude	14.5	The default faintest magnitude for the Target Selection filters. Default value can be changed in the search view.
Automatic backups on change	On	Saves a backup a while after you change your data.
Minutes before backup	60	How long after your last change the automatic backup waits. Each new change restarts the wait.
Backup Folder	Not set	Where backups go. The web version always uses your browser's downloads folder. On the desktop app, until you choose a folder, automatic backups go to your Downloads folder and a manual backup asks where to save.
Backup reminder every	7 days	How long after your last backup Astryx reminds you to make one.

Why this matters

The minimum altitude is the single setting that most affects what Astryx tells you. Near the horizon you're looking through much more air, so stars are dimmer, blurrier, and more washed out by light pollution. A target that's technically up at 15° is rarely worth imaging.

Astryx Tip

35° is a good starting point. If your sky is clean and dark to the south, try 30°. If you image from town, where the bottom of the sky glows, 40° or 45° will keep Astryx from suggesting targets that sit in the murk.

Pick a Theme

The theme list at the top of the sidebar changes Astryx's colors. **Dark** is the default. **Night** turns everything red to protect your night vision at the telescope. **Light** is easier to read indoors, and **Matrix** and **Flat** are there for anyone who likes them.

The list beside it sets the text size, from 8 to 26 pixels. 16 is the default; pick a larger size if the text is hard to read on your screen.

Related

- Introduction: what each view is for.
- Backup and Restore: keeping your locations, equipment, and logs safe.
- Viewfinder: seeing your field of view on a target.

Chapter 3

Your First Night

> **At a glance**
> A clear night is coming and you want to make the most of it. This chapter follows one night from start to finish, through the views you'll use most. Each step points to the chapter that covers that view in full.

Planning a night with Astryx follows the same path every time:

Find → Choose → Check → Frame → Plan → Image → Review

Before you start, make sure your location is chosen in the sidebar (see Getting Started).

1. Find Candidates

Open **Target Selection**. If you already know what you want to image, type its name in the search bar (M 31, Rosette, NGC 7000) and skip to step 3, Check the Night.

Otherwise, let the filters find candidates for you:

1. In **Visibility**, choose the current month. Only targets that are well placed from your location this month remain.
2. In **Type**, choose the kinds of object you like to image, for example emission nebulae and galaxies.
3. Leave **Min Size (′)** and **Limiting Mag** at the values from your Settings, or raise the minimum size if your field of view is wide.

The results update as you change each filter; there's no Apply button.

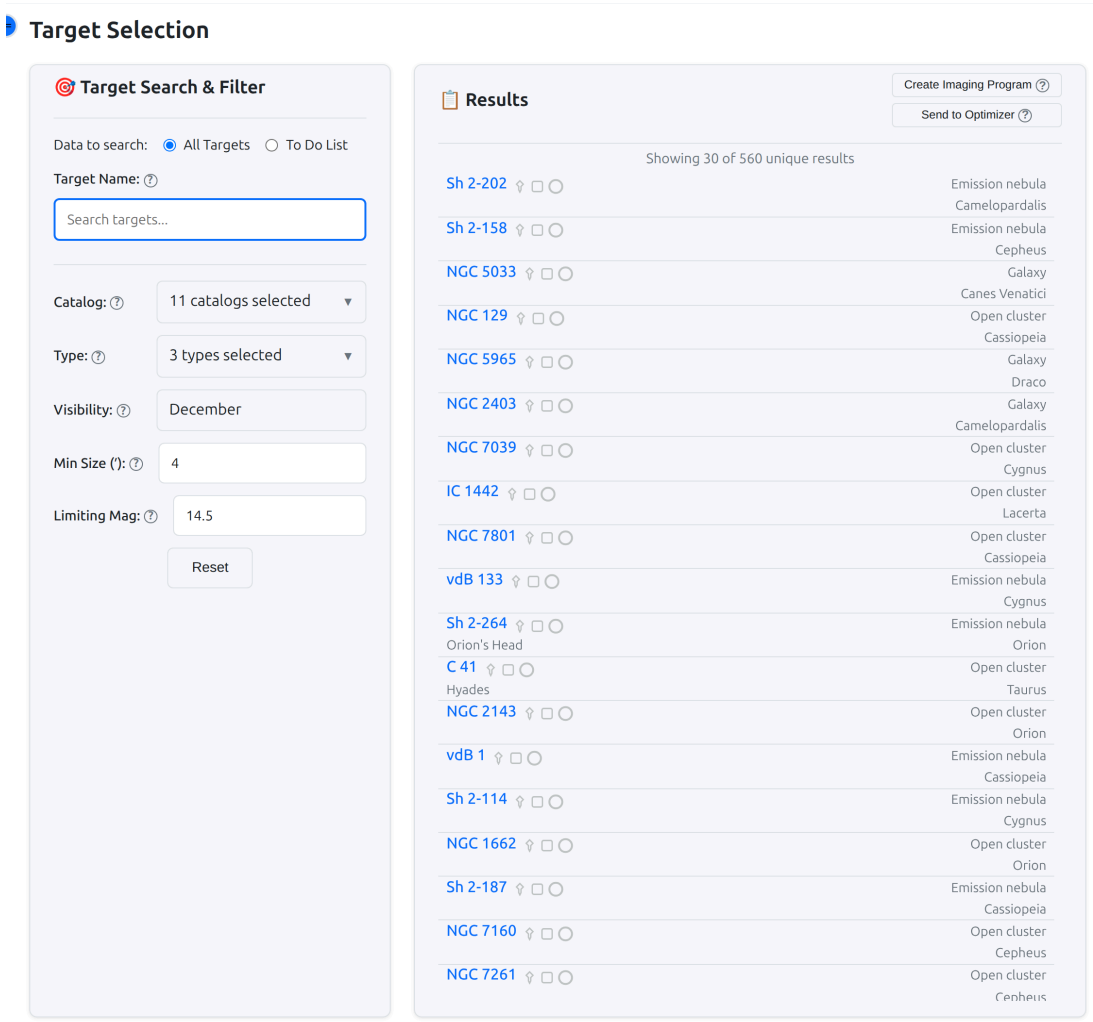


Figure 3: Target Selection with Visibility set to the current month and a few types selected, showing the results list.

> **Why this matters**
> The size filter matters more than it looks. A 2-arcminute galaxy is a smudge a few dozen pixels across on a short refractor. Setting a minimum size that suits your field of view removes targets your setup can't do justice to.

2. Choose Tonight's Targets

A filtered list can still hold hundreds of targets. The Target Optimizer picks the ones that suit *tonight*.

1. In the Results card, click **Send to Optimizer**.
2. Open **Target Optimizer** (see Target Optimizer). The date starts at today.
3. In the source list, choose **Filter Targets**, which now shows how many targets you sent. The results update as soon as you choose.

Each result card shows when the target is above your minimum altitude, how high it climbs, and how far it is from the Moon, with a score for each. The best-scoring targets are at the top.

To image more than one target tonight, click **Best Combinations**. It suggests single targets, pairs, and triplets that fill the night between them. When one looks good, click **Replace Pinned Targets** to pin it.

To image a single target, click **Pin** on its result card.

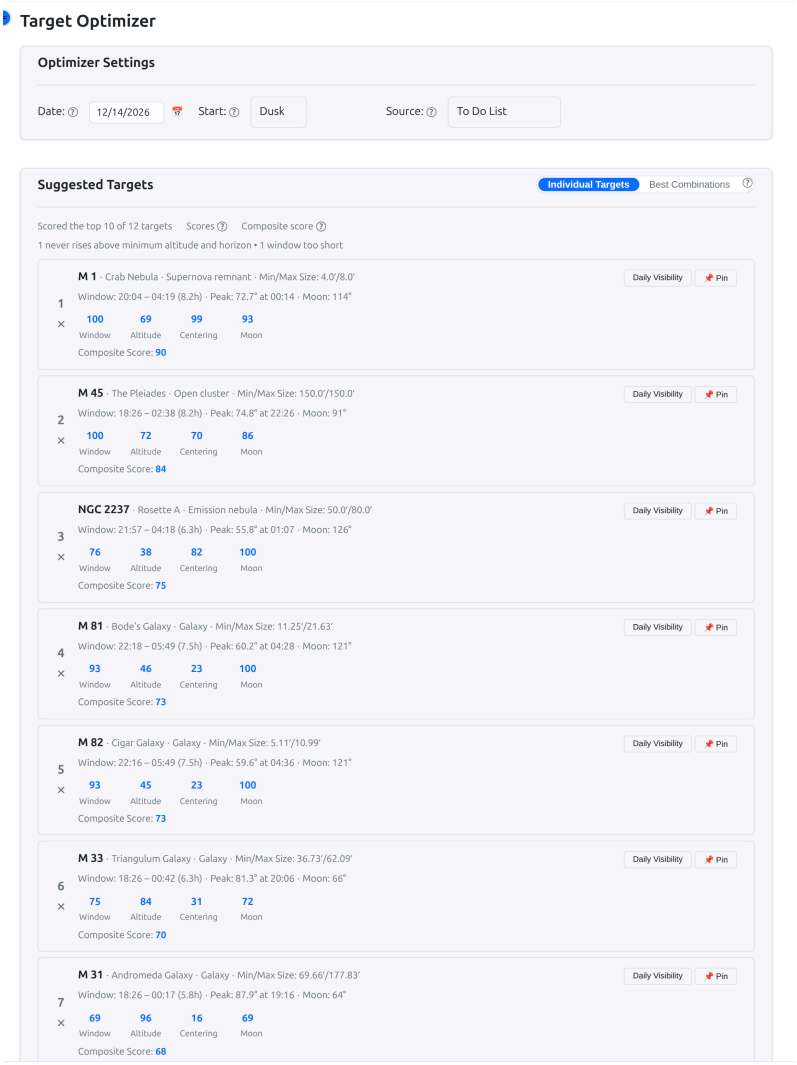


Figure 4: Target Optimizer results, showing the top three cards with their scores.

> **Astryx Tip**
> Use **Add to To Do List** in a target's details whenever you come across something you'd like to image someday. Once your To Do List has a few dozen targets, choose **To Do List** as the Target Optimizer's source instead of filtering each time.

3. Check the Night

For the individual target view, click **Daily Visibility** on a Target Optimizer result card (see Daily Visibility). Or click the target's name under **Pinned Targets** in the sidebar, which makes it the

Current Target and opens its details. Close them, then click **Daily Visibility** in the sidebar. The timeline runs from noon to noon:

- The background shows the sky: white for daylight, gray for twilight or moonlight, black for full darkness.
- The white line is your target's altitude. The dashed line is your minimum altitude, raised where your horizon profile is higher.
- The strips across the top are the cloud, wind, and dew forecast for the day.

The white line is drawn only while the target is above the dashed line, so you can image wherever it shows against a dark background. Below the timeline, the Moon Details and Target-Moon Separation cards tell you how much the Moon will interfere.

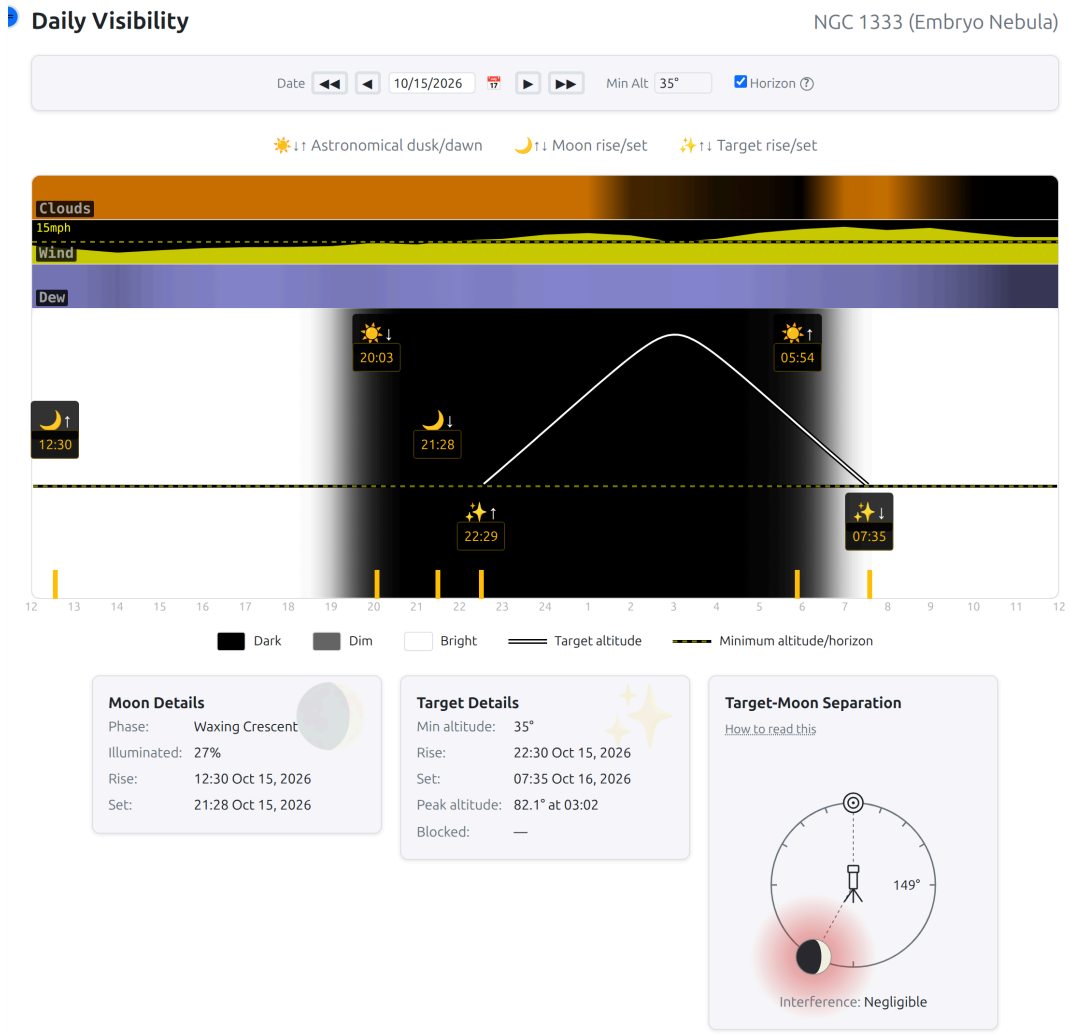


Figure 5: Daily Visibility for NGC 1333 in mid-October: the target clears the dashed line just after the Moon sets and stays well above it until dawn, while the Clouds strip shows cloud coming and going through the night.

4. Frame the Target

Open **Viewfinder** (see Viewfinder), which shows the Current Target. Choose your telescope and sensor, and tick **DSS image**. The frame is your field of view over a real sky survey image. Rotate it to fit the target, or switch to **Wider** and drag the frame to move the center. The Center coordinates below the image tell you where to point the mount.

5. Plan the Night

Open **Sequence Planner** (see Sequence Planner). It builds a plan from your pinned targets as soon as it opens:

1. In **Session Settings**, check the date, location, and start time.
2. Check the other session settings (autofocus, meridian flip, guide calibration, dithering) match how you run your rig. Typically, you will only need to do this once. The Log Analysis can help you with the proper settings.
3. In **Target Allocation**, set each target's exposure length.

The Timeline shows the night with a band for each target. Below it, the Target Sequence in the Imaging Plan lists when to start and stop each target and how many exposures you'll get. Red lines under a target warn of any stretch where it's below your minimum altitude or behind your horizon.

Click a target's name in the Target Sequence to see its predicted session events (autofocus runs, guide calibration, imaging, and the meridian flip) with their start and end times. Click **Download PDF** to keep a copy for the night.

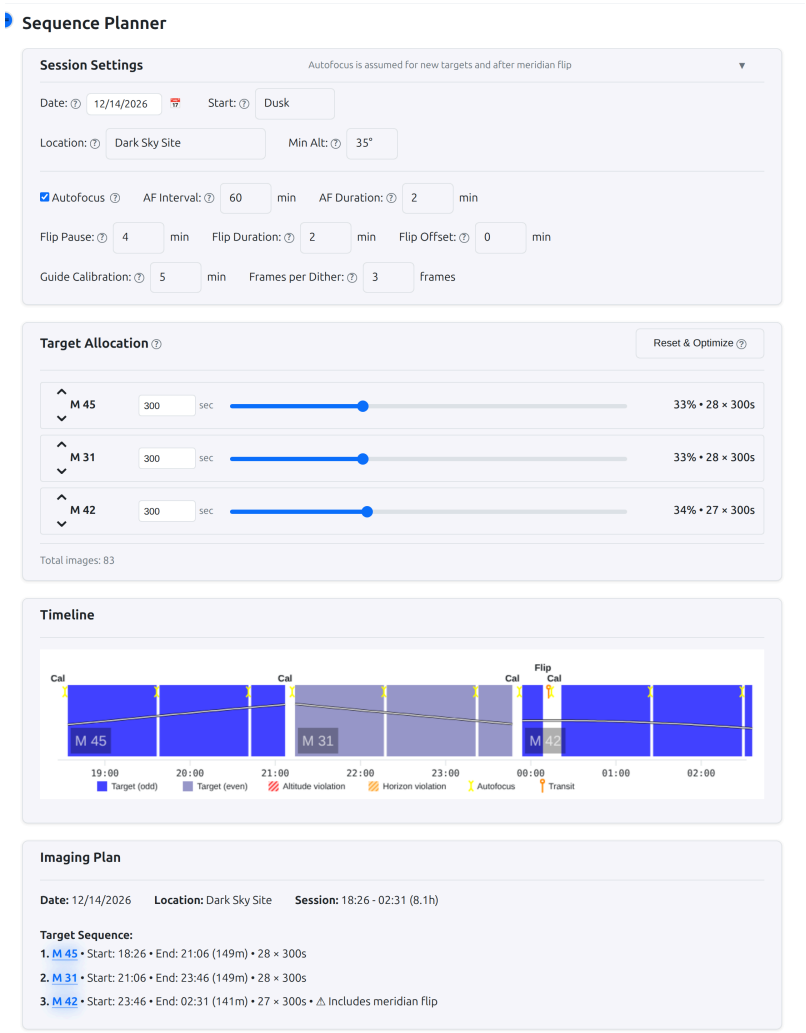


Figure 6: Sequence Planner with three pinned targets: the Timeline and the Imaging Plan.

6. Image

Go image. Astryx doesn't control your equipment; the Imaging Plan is your guide for setting up the sequence in your capture software.

7. Review

As soon as you start imaging, record the night's conditions. After the session, learn from it:

- **Imaging Log.** Create a project for the target (**+ New Project**), then click **+ Add Session** and fill in what you know now: the date, your equipment, the sky conditions, the sub length, and the number of exposures you plan to take. **Calculate Moon Data** fills in the Moon for you. After the session, click the session's row in the project to open it again, and fill in how many exposures you took (**Original Exposures**) and how many you kept (**Used**). Over time the log tracks your total integration on each target, by filter (see Imaging Log).

- **Log Analysis.** After the session is complete, load the night's ASIAir Autorun log and PHD2 guide log (see Log Analysis). The Combined Report tells you how many subs were clean, what went wrong and when, and what to change next time.

> **Why this matters**
> Log Analysis measures how long your camera actually takes between subs and how long each dither takes to settle. The Sequence Planner uses the average of those measurements over your last 20 nights, so its exposure counts match what your rig really does.

Related

- Getting Started: locations, equipment, and settings.
- Finding Targets: searching and filtering.
- To Do List: keeping a shortlist of targets.
- Daily Visibility: reading the night's timeline.
- Viewfinder: framing a target.
- Target Optimizer: ranking targets for a night.
- Sequence Planner: how plans are built.
- Imaging Log: projects, sessions, and programs.
- Log Analysis: reading the night's logs.

Chapter 4

Finding Targets

> **At a glance**
> **Target Selection** is where you find things to image. Search the database by name when you know what you want, or filter it by catalog, type, the month it's well placed, size, and brightness when you're looking for ideas. Click a result to see its details, pin it, add it to your To Do List, or send the whole list to the Target Optimizer.

What You'll See

Open **Target Selection** in the sidebar. The view has two cards:

- **Target Search & Filter**, on the left, is where you say what you're looking for: a name in the search box at the top, or the filters below it.
- **Results**, on the right, lists the matching targets. Clicking one opens its details over the list.

Until you change them, the filters are at their defaults (every catalog and type, any month, and the size and magnitude limits from your Settings), so the Results card already shows a list.

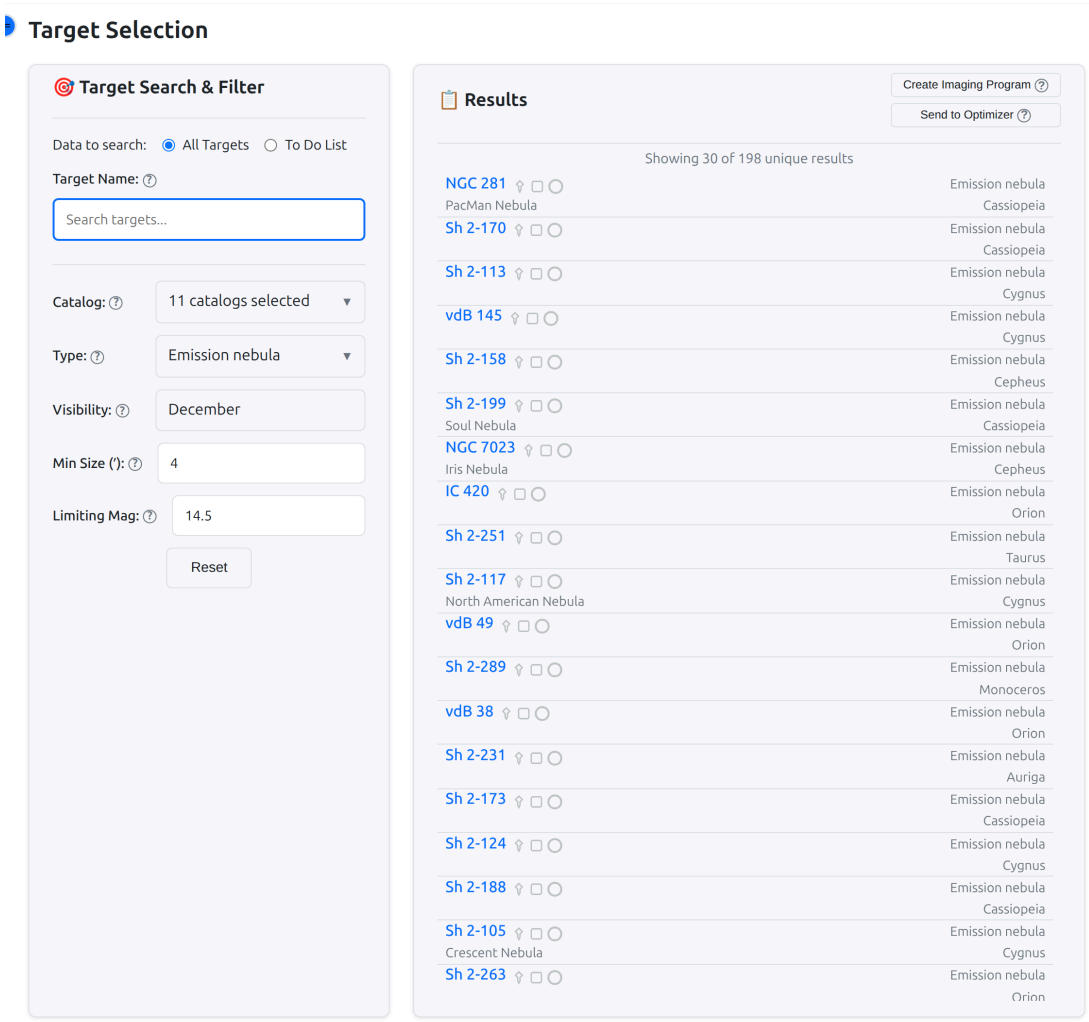


Figure 7: Target Selection with Type set to emission nebulae, Visibility set to the current month, and the Results card showing the list.

Search by Name

Type in **Target Name** at the top of the Target Search & Filter card. Searching starts once you've typed two characters, and the results update as you type.

Astryx looks for what you type in each target's designation, its common names, and its other designations, so all of these work:

- A designation: **M 31**, **NGC 7000**, **Sh 2-155**. You can leave out the space: **M31** finds M 31.
- A common name or part of one: **Rosette**, **Veil**, **Horsehead**.
- A word: **nebula** finds every target with "nebula" in a name.

An exact match comes first, then names that start with what you typed, then the rest.

Starting a search puts the filters back to their defaults, so the search covers every target. Changing a filter clears the search box. Searching and filtering are two ways of filling the same Results list.

> **Astryx Tip**
> **Data to search** above the search box chooses between **All Targets** and your **To Do List**. Switch it to To Do List to find something you saved earlier without wading through the whole database. It works the same way for the filters.

Filter the Database

The filters sit below the search box. Set any combination; the Results card updates each time you change one. There's no Apply button.

Filter	What it keeps
Catalog	Targets from the catalogs you tick. Use **Select All** and **Select None** at the top of the list.
Type	Targets of the types you tick, such as Emission nebula or Galaxy. **Select All** and **Select None** work here too.
Visibility	Targets that are well placed from your location in the month you choose. **Any month** turns this filter off.
Min Size (′)	Targets at least this large, in arcminutes. It starts at the Min Target Size in Settings. Leave it blank to include every size.
Limiting Mag	Targets at least this bright. Lower numbers are brighter. It starts at the Limiting Magnitude in Settings. Leave it blank to include every magnitude.

Reset, below the filters, puts every filter back to its default.

A few things about the filters aren't obvious:

- **An empty Catalog or Type list matches nothing.** If you click **Select None** and forget to tick anything, the Results card is empty.
- **Visibility depends on your location.** A target counts as well placed in a month when, from the location chosen in the sidebar, it spends at least two unbroken hours above your **Min Altitude** (in Settings) in full darkness on nights in that month. Change locations and the list changes with it. With **Any month**, the list includes targets that never rise high enough to image from where you are.
- **Size uses a target's larger dimension.** A 20′ × 3′ edge-on galaxy passes a 10′ minimum.
- **Targets with no magnitude always pass Limiting Mag.** About a third of the database has no magnitude, including most emission and all dark nebulae, so a strict magnitude limit doesn't hide them.

> **Why this matters**
> The database is mostly small galaxies: about 11,400 of its 14,000 targets are galaxies, and most of those are under 3′ across. With every type ticked and no size limit, the list is a wall of faint smudges. Ticking only the types you image and setting a minimum size that suits your field of view turns thousands of results into a few hundred worth considering.

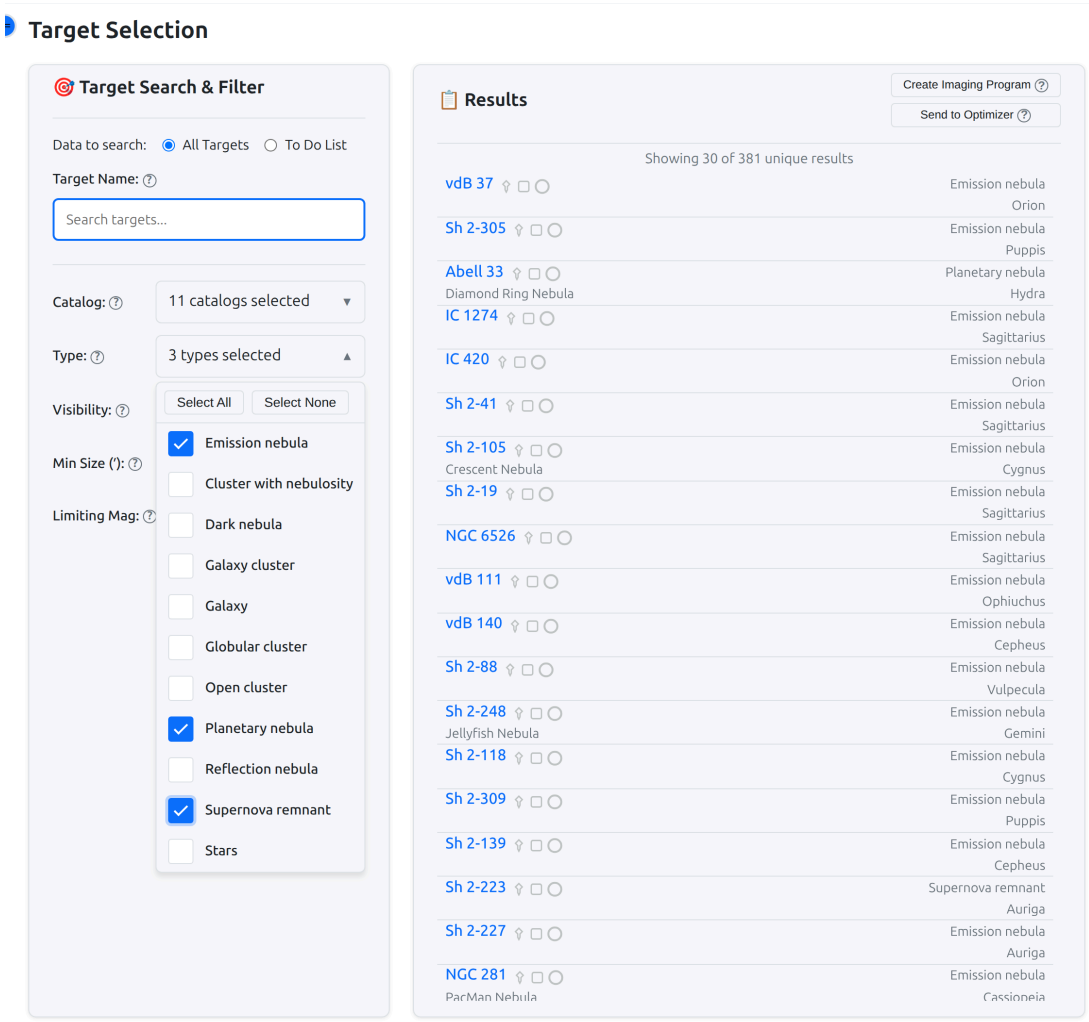


Figure 8: The Target Search & Filter card with the Type list open, showing Select All, Select None, and the ticked types.

Reading the Results

The line above the list says how many targets matched, for example “Showing 15 of 412 unique results”. The list loads more as you scroll.

Each row shows the target’s designation, its type, its common name if it has one, and its constellation. Three small badges sit next to the name:

Badge	Meaning
Pin	Filled when the target is pinned.
Box	Ticked when the target is on your To Do List.
Circle	Empty: no Imaging Log project. Half filled: a project in progress. Filled: a completed project.

A dimmed badge means it doesn’t apply. The badges check every designation a target has, so M 42 shows the same badges as NGC 1976.

> **Note**
> Many objects are in more than one catalog: M 42 is also NGC 1976, and the Cave Nebula is both Sh 2-155 and C 9. When the database links two entries like this, filter results show the object once, under the first catalog in this order: Messier, NGC, IC, Sharpless, Caldwell, Barnard, Arp, Abell. A name search shows every match, so you can find an object under whichever name you know.

Filter results come in a random order, and the order changes each time the filters change. That's deliberate: with hundreds of results, a fixed order would always put the same targets at the top. Search results are in order of how well they match.

Target Details

Click a result and two things happen: the target becomes the Current Target (see Introduction), and its **Target Details** window opens. It's the same window you get by clicking a target anywhere else in Astryx.

The window's header has two buttons:

- **Pin** adds the target to your Pinned Targets in the sidebar. Pin the targets you want to image next; the Sequence Planner plans the night around them. Once it's pinned, the button reads **Unpin**.
- **Add to To Do List** adds it to your To Do List, the longer list of targets you'd like to image someday. Once it's there, the button reads **Remove from To Do List**.

Below the header are four sections:

Section	What it shows
Designation	Name, Type, Common Name, Other Info (other designations), Catalog, and Constellation. The name, type, and common names link to Wikipedia.
Visual Information	Max Size and Min Size in arcminutes, Magnitude, and Surface Brightness. A dash means the database has no value.
Coordinates	Right Ascension and Declination (J2000).
Observability from *your location*	Criteria, Peak Altitude, Best Month, the months it's Observable, and when these were Calculated.

The **Best Month** is when the target is best placed for imaging from your location: high in the sky around midnight, with plenty of dark hours. **Observable** is the run of months it's well placed at all, the same months the Visibility filter uses. If a target never gets high enough, the section says "Not observable with current criteria".

Close the window with **Close** or the Esc key. The results list is still underneath.

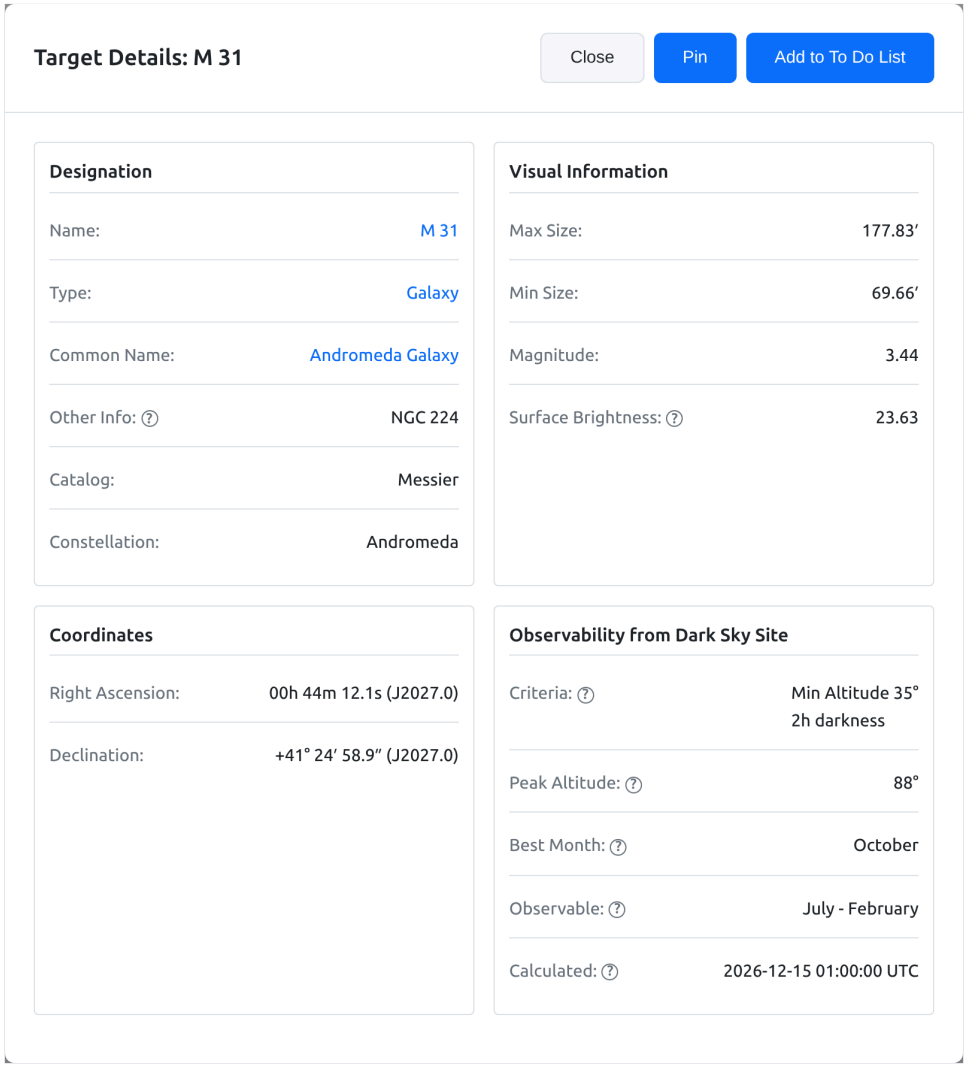


Figure 9: The Target Details window for M 31, showing the Pin and Add to To Do List buttons and the Observability section.

Doing Something With the Results

The two buttons in the Results card's header act on every target in the list, not only the ones you can see. They're hidden when no targets match the filters:

- **Send to Optimizer** hands the list to the Target Optimizer, which picks the best of them for a given night. Open **Target Optimizer** and choose **Filter Targets** as its source. Your First Night walks through this.
- **Create Imaging Program** saves the list as a program in the Imaging Log, for working through a set of targets over many nights. For example, to image every Messier object, click **Reset**, tick only Messier in **Catalog**, and clear **Min Size (′)** and **Limiting Mag**. Then click **Create Imaging Program** and name it "Messier Catalog". Click **Create**; Astryx opens the Imaging Log's Programs tab. A program can hold up to 199 targets, so narrow the filters first if the list is longer.

Tips

> **Astryx Tip**
> Pinned Targets are for tonight; the To Do List is for someday. Add anything that catches your eye to the To Do List as you browse, and pin only what you'll image next. When you sit down to plan, the To Do List gives the Target Optimizer a ready-made list to choose from.

> **Astryx Tip**
> To see what's in season, set **Visibility** to the current month, tick the types you image, and click a few results. The Peak Altitude and Best Month in each target's details tell you whether this month is the start, the middle, or the end of the target's season.

Related

- Your First Night: from a filtered list to a plan for tonight.
- To Do List: your shortlist, and what on it you can image tonight.
- Target Optimizer: ranking a filtered list for a night.
- Getting Started: the Settings that set the starting size and magnitude limits.
- The Target Database: where the targets come from, and what the database holds.
- Best Month Calculation: how the Best Month and the Observable months are worked out.

Chapter 5

To Do List

> **At a glance**
> The **To Do List** is your shortlist: targets you'd like to image someday. It shows which of them you can image on a night and when each one is up, or sorts them by type or by the month they're best placed. Click a target to make it the Current Target.

What You'll See

Open **To Do List** in the sidebar. At the top of the view are three controls:

- **Date:** the night the list is worked out for. It starts at tonight each time you open the view. Every sort uses it: Rise Time and the charts show this night, and Best Month starts at the month before this date. Type a date, click the calendar icon to pick one from a calendar, or use the arrows: ◂ and ▸ move one day, ◂◂ and ▸▸ move a week. Hold an arrow down to keep moving.
- **Sort by:** chooses how the list is arranged: **Rise Time**, **Type**, or **Best Month**. It starts at Rise Time.
- The button next to it switches between a chart and a plain list. It's labeled with the view it switches *to*: **List** while you're looking at the chart, **Chart** while you're looking at the list. The view starts as a chart.

If the list is empty, Astryx says so and points you to Target Selection.

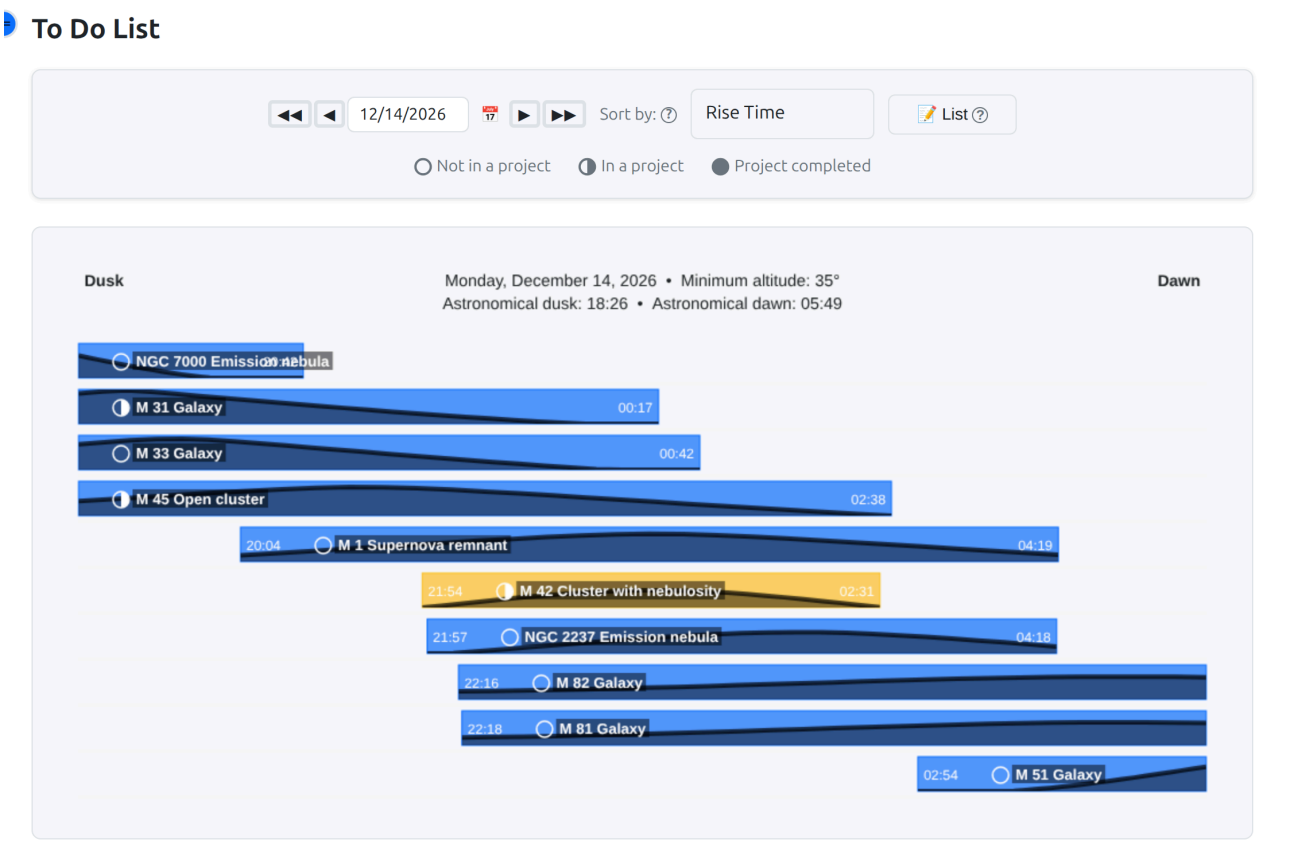


Figure 10: To Do List sorted by Rise Time, chart view: the header with the night's date, Minimum altitude, and dusk and dawn times, and a dozen bars, one of them gold, with a pin before its name, for a pinned target.

Adding and Removing Targets

You add targets from their Target Details window, which opens when you click a target: a result in Target Selection, a name under **Pinned Targets**, or the **Current Target** button (see Finding Targets). Click **Add to To Do List**.

Once the target is on the list, the button reads **Remove from To Do List**. In the To Do List itself, the list view has a **Remove** button on every row. In the chart, click a target's bar to open its Target Details, then click **Remove from To Do List**.

Rise Time

With **Rise Time**, Astryx works out the darkness on the night in the **Date** box, at the location chosen in the sidebar, and splits your list in two:

- **Observable** holds the targets that spend at least two unbroken hours above your minimum altitude between astronomical dusk and dawn. Targets that never drop below your minimum altitude (circumpolar targets) always count.
- **Not Observable** holds the rest.

The minimum altitude here is the **Min Altitude** in Settings.

> **Note**
> This is not the same test as the Visibility filter in Target Selection, which looks at a whole month. A target can be in season by the Visibility filter and still be Not Observable, or the reverse, on a night when it's poorly placed.

The List

In the list view, each card shows its count, for example "Observable (12)". Circumpolar targets come first, then the rest in the order they rise. Each row shows:

- the imaging status circle (empty, half filled, or filled, as in Target Selection),
- a pin and a gold background behind the name if the target is pinned,
- the designation, type, and up to two common names,
- **Rise** and **Set**: the times the target climbs above and sinks below your minimum altitude, in the location's local time. Circumpolar targets show "Circumpolar" for both. If the target is already above your minimum altitude at dusk, Rise is when it climbed above it that afternoon. "N/A" means the target never climbs above your minimum altitude that day.

The Chart

The chart shows only the Observable targets. Its header gives the night's date, your **Minimum altitude**, and the **Astronomical dusk** and **Astronomical dawn** times. The left edge of the chart is dusk and the right edge is dawn.

Each target gets a bar that runs from when it rises above your minimum altitude to when it sets below it, cut off at dusk and dawn:

- A rise time at the left end of the bar means the target rises after dusk. A set time at the right end means it sets before dawn. No time means the bar runs to the edge of the night.
- The line inside the bar is the target's altitude through the night, from the horizon at the bottom of the bar to straight overhead at the top. The peak of the line is when the target is highest.
- A gold bar with before the name is a pinned target.
- The circle before the name is the imaging status.

Click a bar to make that target the Current Target.

> **Astryx Tip**
> Read the chart from left to right to plan a night: a bar ending early in the evening is a target to start with, and one starting after midnight is a target to finish with. The Sequence Planner does this ordering for your pinned targets.

Type and Best Month

The other two sorts group the whole list:

- **Type** makes one group per object type, such as Emission nebula or Galaxy, in alphabetical order. Within a group, targets are in order of their Best Month.

- **Best Month** makes one group per month, starting with the month before the one in the **Date** box, so the targets whose season is ending come first, then that month's, then the months ahead. Targets with no Best Month for your location are in **No Best Month Data** at the end. Typically that's because the target never rises high enough from that location to image, and often never rises above the horizon at all: a far-southern target seen from the north, for example. The Best Month is the one shown in a target's details in Target Selection.

The list view shows every target in your To Do List, with the number of targets in each month. The chart view shows the same groups, but only the targets that are Observable on the night in the Date box, so it can leave out whole groups.

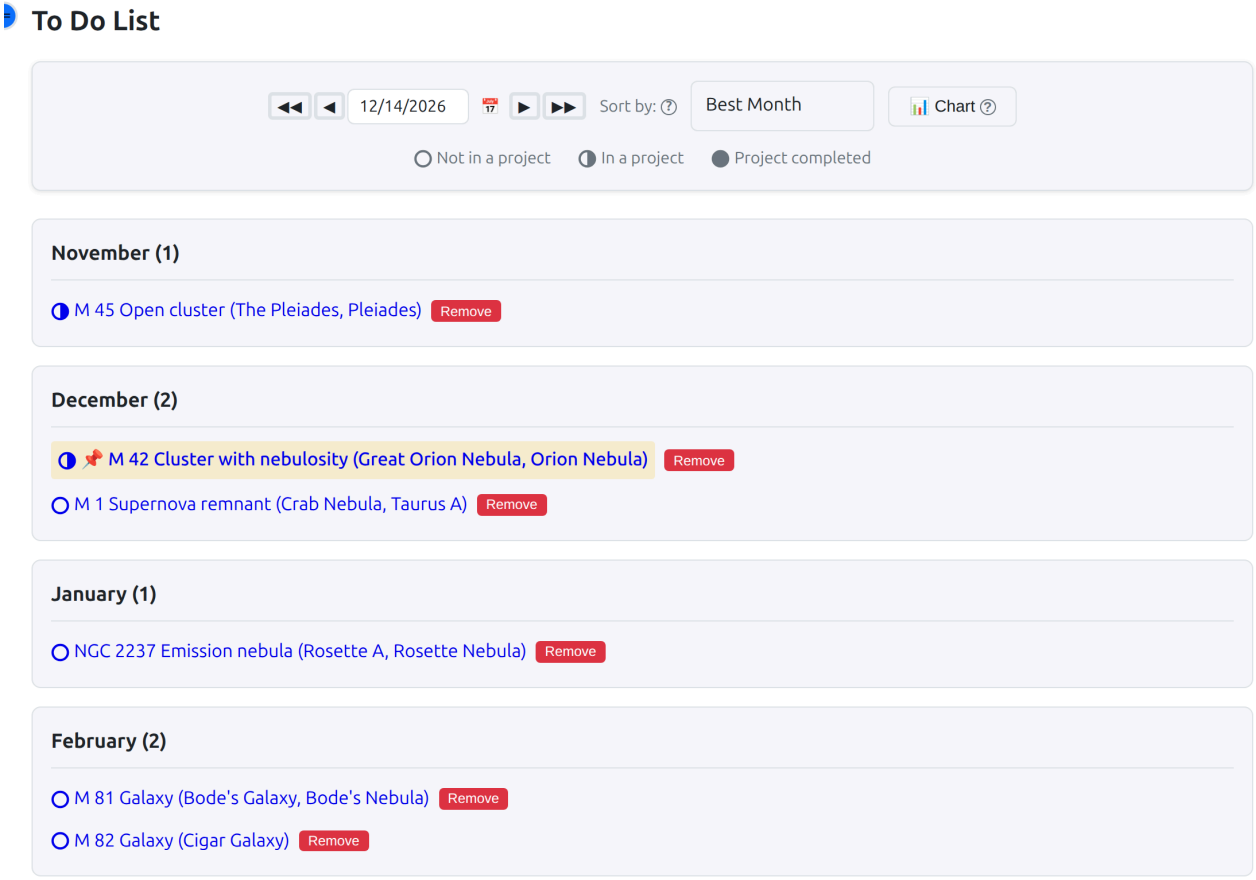


Figure 11: To Do List sorted by Best Month, list view: cards for last month, this month, and the next two months, with Remove buttons.

Clicking a Target

Click a target's name in the list, or its bar in the chart, and Astryx:

- makes it the Current Target, so Daily Visibility, Yearly Observability, and the Viewfinder show it, and
- opens its Target Details window, with the target's designations, size, coordinates, and observability.

Close the details window and open **Daily Visibility** to check the night for that target.

Tips

> **Astryx Tip**
> Use the To Do List as the Target Optimizer's source. Choose **To Do List** in the Target Optimizer's source list: it ranks everything on your list for the night you choose, without filtering the whole database again.

> **Astryx Tip**
> In Target Selection, set **Data to search** to **To Do List** to search or filter only your To Do List, for example to see which of your saved targets are galaxies over 10′.

> **Why this matters**
> The To Do List is for someday; Pinned Targets are for tonight. Keep the To Do List long and the pinned list short. The Sequence Planner builds its plan from your pinned targets only.

Related

- Finding Targets: searching, filtering, and adding targets.
- Your First Night: from a list of targets to a plan for tonight.
- Target Optimizer: ranking your list for a night.
- Getting Started: the Min Altitude setting.
- Best Month Calculation: how each target's Best Month is worked out.

Chapter 6

Yearly Observability

> **At a glance**
> **Yearly Observability** shows a whole year for one target: how high it gets during full darkness each night, and how good each night is for imaging it. Use it to find a target's season, see where the full Moons fall in it, and understand why a target isn't available in a given month.

What You'll See

Open **Yearly Observability** in the sidebar. It shows the Current Target from the location chosen in the sidebar (see Introduction). To look at a different target, change the Current Target, for example by clicking it in Target Selection or the To Do List, then come back. Changing the location in the sidebar redraws the graph.

At the top are the target's name and common name, and a summary line:

- **Peak altitude**: the highest the target gets during astronomical darkness on any night of the year.
- **Best month** and **Observable**: the same Best Month and run of Observable months shown in the target's details in Target Selection. They're missing if the target has no Best Month from this location.

Below the summary, **Minimum Altitude:** sets the altitude you need, from 5° to 60°. It starts at the Min Altitude in Settings.

Under the graph, a legend names each part of it: Target Altitude, Minimum Altitude, Observability (darker = better), Full Moon, and Current Day.

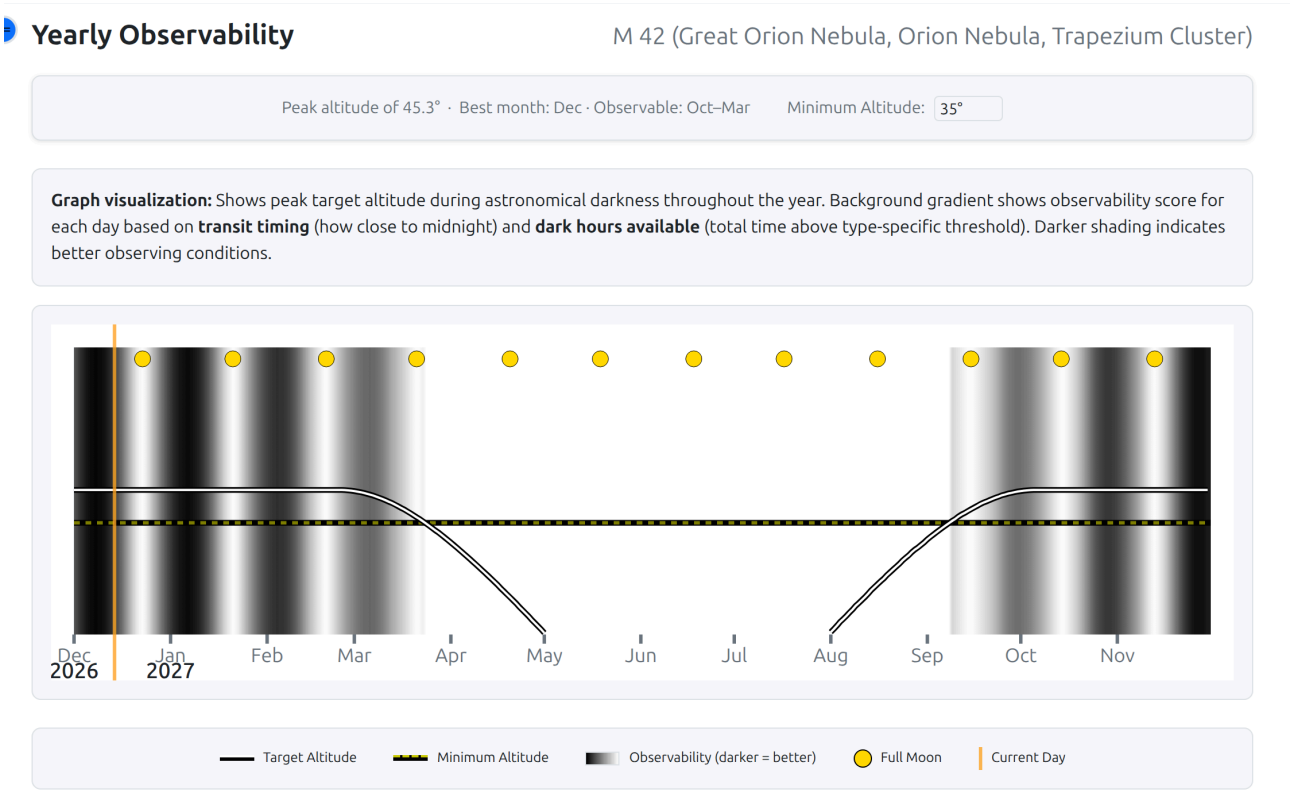


Figure 12: Yearly Observability for M 42 from a mid-northern location: the summary line, the Minimum Altitude control, and the graph with its dark season and the full-Moon markers.

Reading the Graph

The graph covers twelve months, starting on the first day of the current month. Month names run along the bottom, with the year under the first month and under each January. Height on the graph is altitude, from the horizon at the bottom to straight overhead at the top.

Part	What it shows
White line	The target's highest altitude during astronomical darkness on each night. Gaps mean it stays below the horizon all night.
Yellow dashed line	Your Minimum Altitude.
Background	How good each night is for imaging this target. Darker is better. White means the target doesn't reach your Minimum Altitude in darkness that night, or there's no astronomical darkness at all.
Gold circles along the top	Full Moons.
Orange vertical line	Today.

The background shading takes three things into account for each night:

- how close to midnight the target crosses the meridian (its highest point),
- how many hours it spends high enough in full darkness, and
- the Moon: how bright it is, and how close it comes to the target.

So the dark stretch of the background is the target's season, and the pale gaps that recur inside it, centered on the gold circles, are the full-Moon weeks.

> **Note**
> The Minimum Altitude here only moves the dashed line and turns nights below it white. It doesn't change how the remaining nights are shaded. That shading uses a fixed altitude for each type of target; Yearly Observability Details has the values.

The Shape of the Line

The white line is flat across the middle of a target's season, and slopes down on either side. That's because of *when* the target is highest, not how high it can get:

- **Flat:** every night, the target reaches the same highest altitude when it crosses the meridian. When that happens during full darkness, the line shows the full altitude, so it stays level.
- **Sloping:** the target crosses the meridian about four minutes earlier each night, or two hours earlier each month. Before its season, it's highest after dawn; after its season, it's highest before dusk. Either way, darkness only catches it while it's lower, so the line slopes down.

How long the flat stretch lasts depends on how long your nights are: long winter nights catch more of each target's arc than short summer ones.

> **Why this matters**
> The line shows how high a target *can* get, not how long it stays there. Early and late in the season, a target may reach a good altitude for only an hour at the start or end of the night. Check those nights in Daily Visibility before you plan around them.

Using It

- **Find the season.** Look for the dark stretch of background and the flat part of the line. That's when to image the target.
- **Avoid the Moon.** Within the season, plan your nights in the dark gaps between the gold circles.
- **See why a target isn't available.** If the line never reaches the dashed line, the target never gets high enough from this location. If it does, but not this month, it's out of season; the graph shows when it comes back.
- **Set the bar.** Raise **Minimum Altitude** if trees or buildings block your low sky, or if you want the steadier air higher up. The white parts of the background grow to show the nights you'd lose.

Tips

> **Astryx Tip**
> Check Yearly Observability before starting a long project. A target that needs twenty hours of integration needs a season with enough dark, moon-free nights to collect them. If the season is nearly over, start something else and come back next year.

> **Astryx Tip**
> Set **Minimum Altitude** to match the Min Altitude in Settings, so the dashed line here is the same one you see in Daily Visibility and the Sequence Planner.

Related

- Finding Targets: the Best Month and Observable months in a target's details.
- To Do List: changing the Current Target from your shortlist.
- Yearly Observability Details: how the shading is scored.
- Best Month Calculation: how the Best Month and Observable months are worked out.

Chapter 7

Daily Visibility

> **At a glance**
> **Daily Visibility** shows one night for one target, from noon to noon: when it's dark, when the target is high enough and clear of your horizon, where the Moon is, and the cloud, wind, and dew forecast. Use it to decide whether a night is worth setting up for.

What You'll See

Open **Daily Visibility** in the sidebar. It shows the Current Target (see Introduction) from the location chosen in the sidebar, and the target's name is next to the view's title. To check a different target, change the Current Target, for example by clicking **Daily Visibility** on a Target Optimizer result card. Changing the location in the sidebar redraws the view.

From top to bottom, the view has:

- the controls: **Date**, **Min Alt**, and **Horizon**,
- the forecast strips: **Clouds**, **Wind**, and **Dew**,
- the timeline, with the times of the night's events and the target's altitude, and
- three cards: **Moon Details**, **Target Details**, and **Target-Moon Separation**.

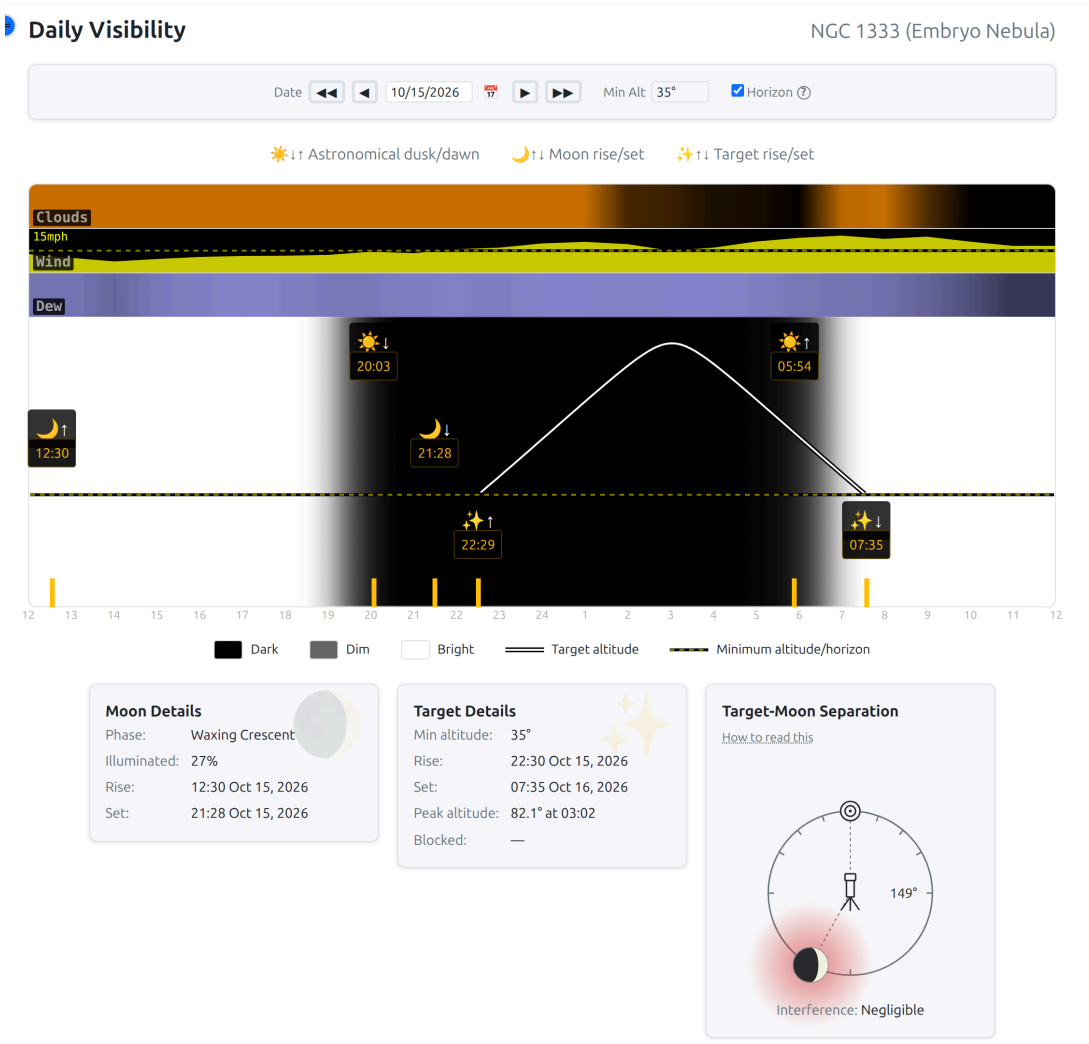


Figure 13: Daily Visibility for NGC 1333 in mid-October: the controls, the three forecast strips, the timeline with its markers and the target's altitude line from moonset to dawn, and the three cards below.

The Controls

Control	What it does
Date	The night to show: the evening of this date and the morning after. Type a date in your Date Format (see Getting Started), click the calendar icon to pick one from a calendar, or use the arrows: ◂ and ▸ move one day, ◂◂ and ▸▸ move a week. Hold an arrow down to animate back and forth through time.
Min Alt	The lowest altitude that counts, from 5° to 60°. It starts at the Min Altitude in Settings.
Horizon	When ticked, your location's horizon profile counts too, so the target must clear both. Untick it to see the night as if your horizon were flat.

The date starts at today. After midnight but before noon it starts at yesterday instead, so in the small hours you see the night you're in, not the coming one.

The Timeline

The timeline runs from noon on the date you chose to noon the next day. The hours are labeled along the top, in your location's local time. On the night the clocks change, an hour repeats (autumn) or is skipped (spring), so each label stays over its clock time.

The Sky

The background shows how dark the sky is, judged from the Sun and the Moon:

- **Bright** (white): daylight, or a bright Moon high in the sky.
- **Dim** (gray): twilight, or moonlight.
- **Dark** (black): full darkness, with the Moon down or too faint to matter.

The Moon's effect depends on its phase, how high it is, and how close it is to the target, so a crescent Moon low in the west barely dims the sky, while a full Moon near the target washes out most of the night.

The Lines

- The **white line** is the target's altitude, from the horizon at the bottom to straight overhead at the top. It's drawn only while the target is above your minimum altitude and clear of your horizon, so wherever you see it, the target is up.
- The **yellow dashed line** is your minimum altitude. With **Horizon** ticked, it rises wherever your horizon profile is higher than the minimum altitude, at the times the target passes behind it.

You can image while the white line is visible against a dark background.

The Markers

Small yellow ticks along the bottom of the timeline mark the night's events, each with its time. They're in three rows:

Row	Markers
Top	Astronomical dusk and dawn: the Sun 18° below the horizon.
Middle	Moonrise and moonset.
Bottom	When the target rises above, and sets below, your minimum altitude and horizon.

A marker is missing when the event falls outside the noon-to-noon window, or doesn't happen that day.

The Forecast Strips

The three strips above the timeline are the weather forecast for the same noon-to-noon window, hour by hour, from Open-Meteo:

Strip	What it shows
Clouds	Total cloud cover. Black is clear; bright amber is overcast.

Strip	What it shows
Wind	Wind speed 10 m above the ground, as a yellow graph. The number at the top left is the top of the scale in mph, and the dashed line is half of it. Sheltered sites may get less wind than this.
Dew	The risk of dew, from how close the air temperature is to the dew point. Black is safe; bright blue means dew is likely.

The forecast covers today and the next four days, plus last night until noon. For other dates, the strips are replaced by “Cloud forecast unavailable for this date”. Daily Visibility needs an internet connection for the forecast; everything else works offline.

> **Astryx Tip**
> Hold the pointer over a strip for a reminder of what it shows.

The Cards

Moon Details

The Moon’s **Phase**, how much of it is **Illuminated** during the night, and when it will **Rise** and **Set**. Rise and Set are listed in time order, each with its date, so the Moon often sets before it rises. When the Moon rises in the noon-to-noon window but sets just after it (or set just before it rose), the card still shows that time, but the timeline doesn’t. If the Moon neither rises nor sets in the window, the card says “No rise/set during observation”.

Target Details

Line	What it means
Min altitude	The Min Alt you chose.
Rise, Set	When the target rises above and sets below your minimum altitude and horizon. If the target is already up at dusk, Rise is when it came up that afternoon; if it’s still up at dawn, Set is when it goes down that morning. “Before dusk” and “After dawn” mean it stays up all day and never sets. “Never rises” means it doesn’t climb above your minimum altitude and horizon at all that day.
Dips below min	Shown only when the target drops below the dashed line during the night and comes back up, usually behind a tree or building in your horizon profile.
Peak altitude	The highest the target gets during darkness, and when.
Blocked	How many minutes the target is above your minimum altitude but behind your horizon profile. A dash means none.

Target-Moon Separation

A dial shows where the Moon is in relation to your target:

- **You** are the telescope in the center.
- **The target** is the bullseye at 12 o’clock.

- **The Moon** is on the rim, drawn in its phase. On the left it's approaching the target; on the right it has passed it. Near 12 o'clock it's at its closest, and at 6 o'clock it's on the far side of the sky.
- **The number** beside the dial is how far apart the Moon and the target are, in degrees: the closer of the separations at the start and end of the target's night.

The Moon moves about 13° a day against the stars, so as you step through the nights with ▸, it travels clockwise round the dial, once a month. Many targets lie well away from the Moon's path, and the Moon passes beside them rather than over them. For these, it lifts outside the rim as it passes, and arcs over the bullseye. For the same reason, the Moon at 6 o'clock is never a full 180° from such a target, only 180° less the target's distance from the Moon's path: about 150° for M 42, and about 130° for M 51.

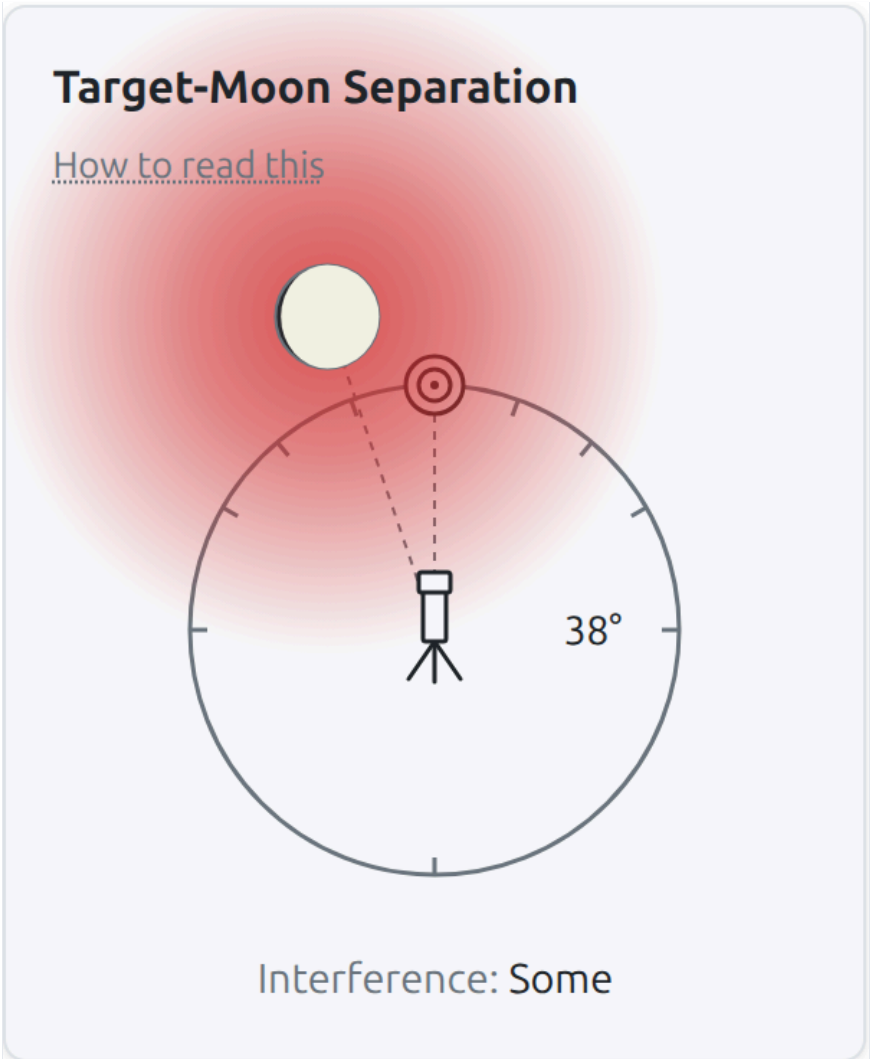


Figure 14: The Target-Moon Separation card on a night with a bright Moon near the target: the telescope in the center, the bullseye at the top, the Moon on the rim with its haze reaching the target, the separation in degrees, and the Interference line below.

The haze around the Moon is its light in your sky. The brighter the Moon, the further the haze spreads: a full Moon's reaches 90°, a half Moon's about 50°, and a thin crescent's hardly at all. The deeper the target sits in the haze, the brighter the sky behind it.

If the Moon stays below the horizon from dusk to dawn, it's drawn as an outline, with no haze.

Interference, below the dial, sums it up. For a full Moon it goes by separation:

Separation, full Moon	Interference
Under 20°	Severe
20° to 40°	Significant
40° to 60°	Some
60° to 90°	Minimal
90° or more	Negligible

A fainter Moon interferes less at the same separation, in proportion to how far its haze reaches. A half Moon 30° from the target rates as a full Moon would at about 50°: **Some**, not **Significant**. Interference is **Negligible** whenever the haze doesn't reach the target, and when the Moon stays below the horizon all night.

> **Why this matters**
> The Moon's phase alone doesn't tell you whether a night is usable. A bright Moon on the far side of the sky, or one that sets early, can still leave hours of good imaging. Read the separation together with the timeline: if the background is dark while the white line is up, the Moon isn't a problem, whatever its phase.
> It also depends on the target. Bright emission nebulae, especially imaged with narrowband filters, and globular clusters can be good targets even with a bright Moon. Save the dark nights for faint galaxies and reflection nebulae.

Reading the Night

A good night has all of these:

- a long stretch of black background,
- the white line visible through most of it,
- black Clouds and Dew strips over the same hours, and
- a low Wind graph.

Then look at the details: when the target rises and sets tells you when to start and stop, and Blocked tells you how much of the night your horizon costs you.

Tips

> **Astryx Tip**
> Step through the nights with ▸ to find the next good one. The timeline and the cards update each time, and the forecast strips cover the next four days.

> **Astryx Tip**
> If **Blocked** shows a lot of minutes, untick **Horizon** to compare. If the target would be well placed on a flat horizon, it may be worth imaging from a different spot, or a different location.

Related

- Your First Night: checking a night as part of planning it.
- Getting Started: horizon profiles and the Min Altitude setting.
- Yearly Observability: the same target over the whole year.

Chapter 8

Viewfinder

> **At a glance**
> The **Viewfinder** shows how the Current Target fits your telescope and camera, over a real image of the sky. Turn the camera, move it to a better center, and note the angle and coordinates to use at the telescope.

What You'll See

Open **Viewfinder** in the sidebar. It shows the Current Target (see Introduction). The view has three rows.

At the top:

- **Current Target**: the target's designation, with its common name in parentheses (**Object**), and its **Size** in arcminutes. A name too long for the card ends in "…"; point at it to see all of it.
- **Equipment**: the **Telescope:** and **Sensor:** lists.

In the middle is the **Field of View** card. On its left are the **Actual** / **Wider** switch; four checkboxes: **DSS image**, **Target size**, **Full moon**, and **Crosshair**; the **Rotate camera to:** control; the coordinates of the frame's center; and the **Snapshot** button. The rest of the card is the image of the sky with your frame on it.

At the bottom is the results card: the numbers for your telescope and sensor together.

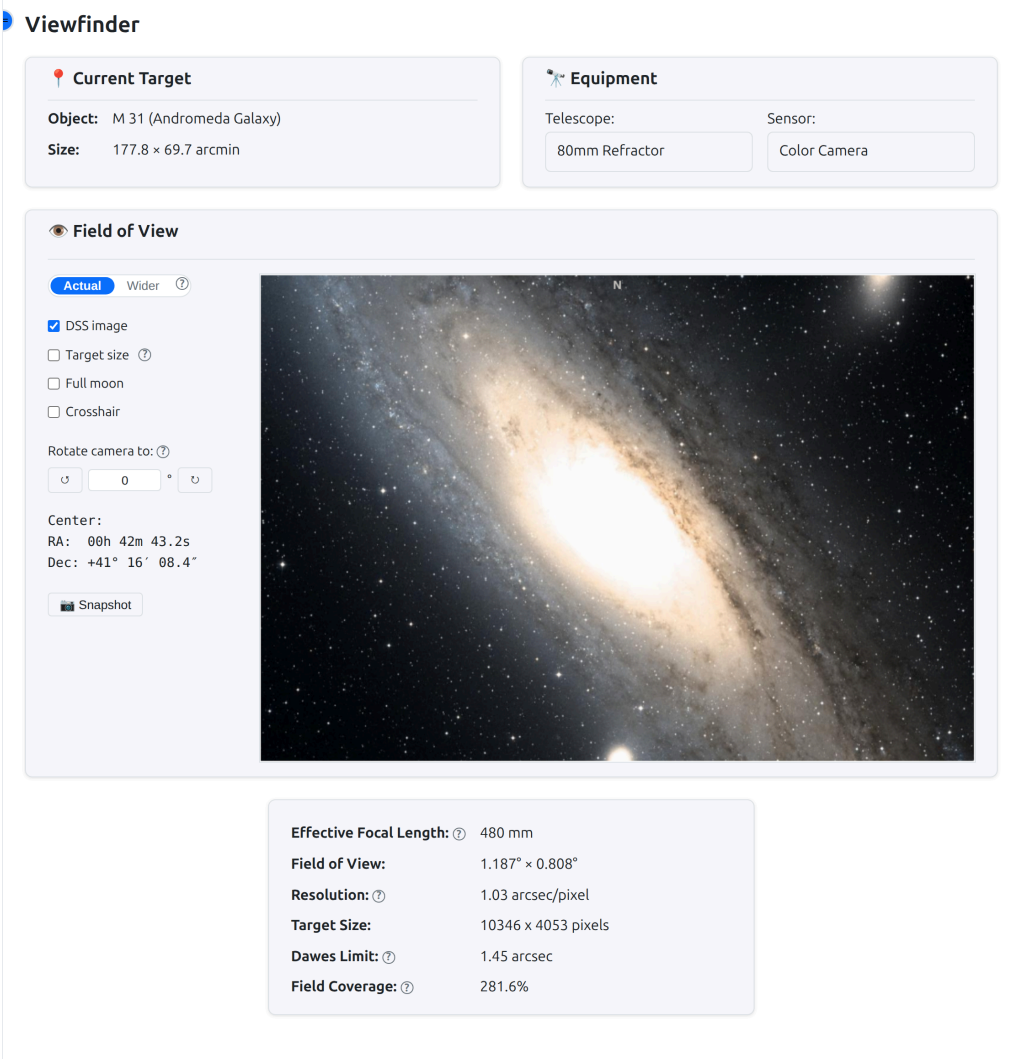


Figure 15: The Viewfinder for M 31 with a telescope and sensor chosen and DSS image ticked: Current Target and Equipment at the top, the Field of View card with its controls beside the sky image, and the results below.

Choosing Your Equipment

Choose a telescope from **Telescope:** and a camera from **Sensor:**. The lists hold the equipment you saved in Admin Tools (see Getting Started). Astryx remembers your choice, so the next time you open the Viewfinder it's already set.

Until both are chosen, the results card says "Select telescope and sensor to calculate field of view."

The Results

Line	What it means
Effective Focal Length	The telescope's focal length times its reducer or Barlow multiplier, in mm.
Field of View	How much sky the sensor covers, width × height, in degrees.

Line	What it means
Resolution	How much sky each pixel covers, in arcseconds per pixel.
Target Size	About how many pixels the target spans on your sensor, from its catalog size.
Dawes Limit	The finest detail the telescope's aperture can separate, in arcseconds (116 ÷ aperture in mm).
Field Coverage	How much of the field's area the target fills, as the oval or circle drawn for it. Shown only when the target's size is known.

If the target is large for your field, Astryx tells you so: when it spans more than 70% of the field's shorter side, or fills more than half its area, a message suggests an effective focal length that would leave more room around it. The message appears once for each combination of target, telescope, and sensor.

> **Why this matters**
> Resolution compares your camera with the sky's steadiness. On a typical night, the stars are blurred to 2 to 4 arcseconds, so 1 to 2 arcseconds per pixel records all the detail the sky allows. Much finer than that, and you gain nothing but larger files and fainter pixels. Much coarser, and the stars start to look square.

The Sky Image

With **DSS image** ticked, Astryx downloads a color image of the target's part of the sky from the Digitized Sky Survey (DSS2), through the CDS image service in Strasbourg, and shows it in your frame. The frame has your sensor's shape, long side across, and fills the width of the card. N marks north: at the top until you turn the camera.

Images are kept on your computer for a while, so a target you've looked at recently appears at once and works offline. An image you don't use again within 15 days is deleted, and a Wider image within 7 days; the next time you need it, Astryx downloads it again. Fetching a new one needs an internet connection; if it fails, Astryx says "Could not load DSS background image — check your internet connection."

Without the image, the canvas is black and shows only the frame and the **Target size**, **Full moon**, and **Crosshair** overlays. **Wider** needs the image.

The Overlays

Checkbox	What it draws
Target size	A yellow dashed ellipse the target's catalog size. It's always drawn tilted at 45°; only its size is meaningful, not its angle.
Full moon	A circle the size of the full Moon (31′), for a sense of scale. The note "Full moon outlined" appears at the bottom of the image.
Crosshair	A small cross at the center of the frame.

The view starts with **DSS image** ticked and the other three unticked.

Rotating the Camera

Rotate camera to: turns the camera the way you'd turn it on the focuser. In Actual, the frame stays put and the sky turns inside it, as it would on the camera's screen; the N marker moves with the sky. Type an angle in degrees, or click ↺ and ↻ to turn the camera one degree at a time; hold a button down to keep turning. ↻ turns the camera clockwise, so the sky turns counterclockwise.

Use it to fit a long target along the diagonal or the long side of the sensor, or to keep a bright star out of a corner. The angle starts at 0° (north up) each time you open the view.

Moving the Center: Wider

The **Actual** / **Wider** switch at the top of the Field of View controls changes what the canvas shows:

- **Actual** shows your field, centered on the target, or on the center you chose in Wider.
- **Wider** shows a sky image three times as wide, centered on the same center as Actual, with your field as a dashed frame you can drag. The pointer becomes a hand over the frame. The **Center:** coordinates beside the image follow the frame as you drag.

To move the center:

1. Click **Wider**.
2. Drag the frame to where you want it, and rotate it if you like.
3. Click **Actual**. Astryx fetches a new image centered where you left the frame.

To keep going, click **Wider** again: Astryx fetches a new wider image around the new center, and you can drag on from there. Step by step, you can walk the frame as far across the sky as you like.

Once you've moved the frame, its center is no longer the target, so the Current Target card goes blank, **Target size** is turned off and can't be ticked, and the results card drops **Target Size** and **Field Coverage**. The **Center:** coordinates always show the center of what you're looking at. Choose a target again (from the Pinned Targets, for example) to start over from it; leaving the Viewfinder starts over too.

Target size and **Full moon** are turned off in Wider. **Full moon** comes back when you return to Actual, and so does **Target size** if you haven't moved the frame.

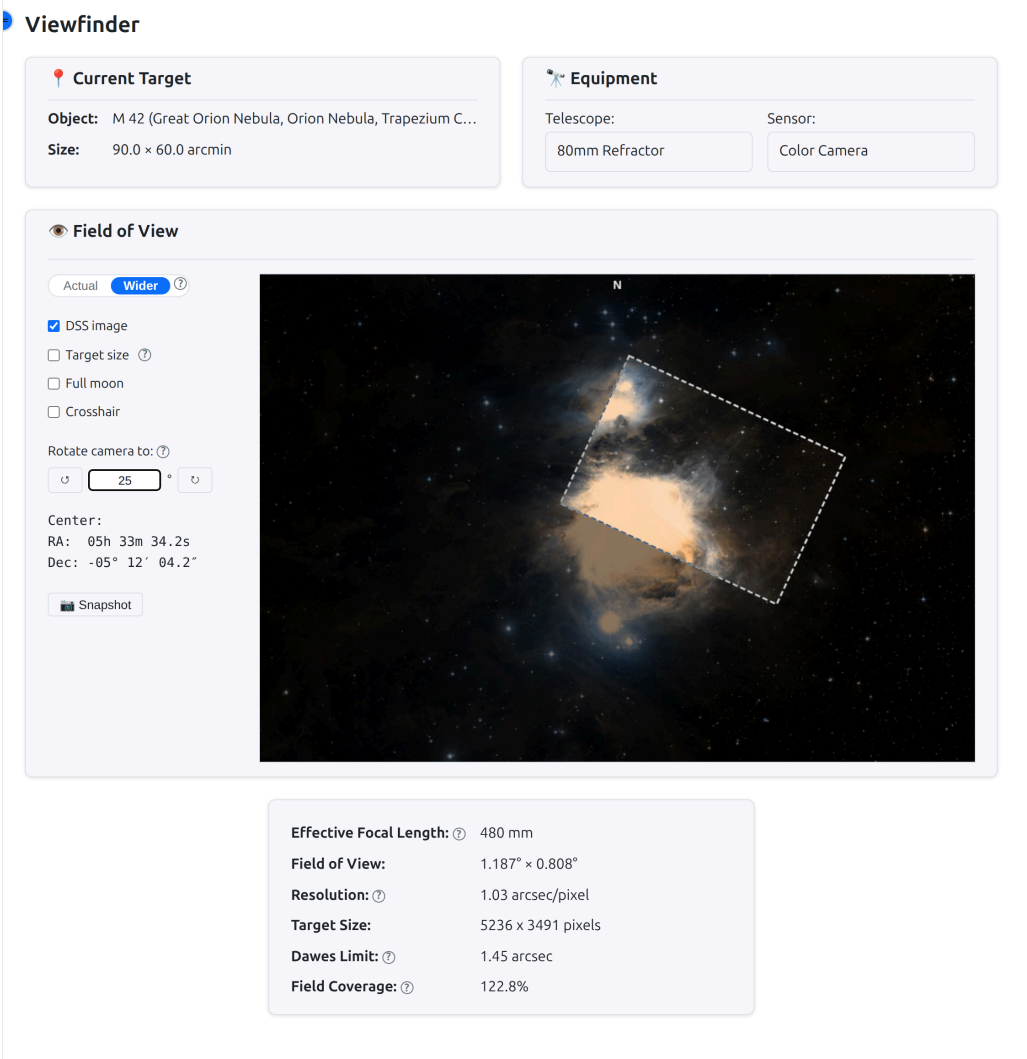


Figure 16: The Viewfinder in Wider mode: the larger sky image, the dashed frame moved off-center and rotated, and the Center coordinates beside it.

The Center Coordinates

The **Center:** coordinates beside the image, for example 05h 35m 17.3s and −05° 23′ 28.0″, are the right ascension and declination of the frame's center. With the rotation angle, they're everything you need to reproduce the framing at the telescope: enter it as the target position in your capture software, and plate solving centers the mount on it.

Snapshot

Snapshot shows the framing on its own, as the camera would record it. In Actual, it's the whole canvas. In Wider, it's just what's inside the frame, turned upright. To keep it, right-click the image and choose the save image option. Click **Close**, or click outside the image, to put it away.

Tips

> **Astryx Tip**
> Check a target in the Viewfinder before you add it to your plan. If it overflows the frame, or is a speck in the middle, another telescope may suit it better, or it may need a mosaic.

> **Astryx Tip**
> A target listed with no size in the database shows “Unknown” in the Current Target card. The sky image still shows it as it is; use that to judge the fit.

Related

- Getting Started: adding telescopes and sensors.
- Your First Night: framing a target as part of planning a night.
- Finding Targets: choosing the Current Target.

Chapter 9

Target Optimizer

> **At a glance**
> The **Target Optimizer** takes a list of targets, your To Do List or the results of a filter, and ranks them for one night: how long each is up, how high it climbs, whether it's best in the middle of the night, and how much the Moon gets in the way. It can also suggest two or three targets that share the night well.

What You'll See

Open **Target Optimizer** in the sidebar. The **Optimizer Settings** card at the top holds the controls. The results appear below it, in the **Suggested Targets** card, and update whenever you change the Date, Start, or Source; there's no button to press.

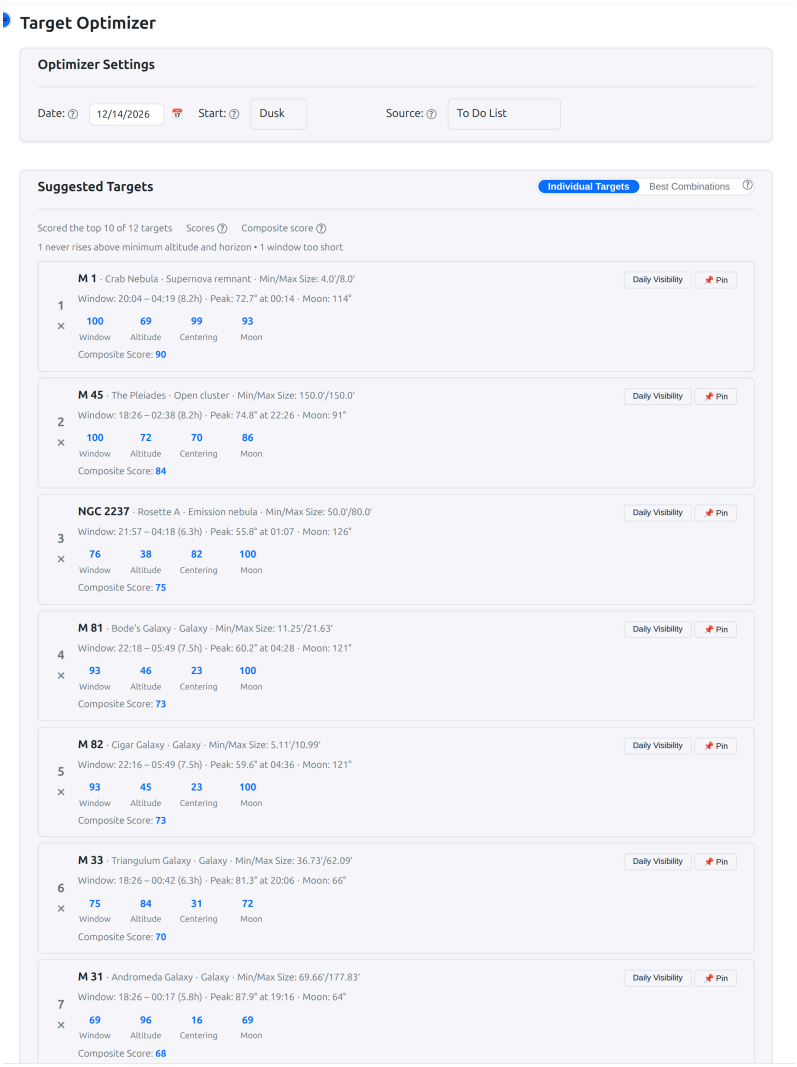


Figure 17: The Target Optimizer with results: the Optimizer Settings card, the summary line, and the first three result cards.

The Settings

Control	What it does
Date	The night to plan: the evening of this date and the morning after. It starts at today.
Start	**Dusk** starts the night at astronomical dusk. **Custom** shows a time field for your own start, for example if you can't be at the telescope until 23:00. It starts out at dusk; type the time as 24-hour HH:MM (2300 works too).
Source	The targets to rank: **To Do List**, or **Filter Targets**, the targets you sent from Target Selection.

The night always ends at astronomical dawn. The Optimizer uses the location chosen in the sidebar and the Min Altitude from Settings.

When you come back to the Optimizer, it shows the results as you left them, with any cards you removed still gone. If you've changed the location or the Min Altitude since, it ranks the targets again for the new ones.

A Custom time is in your location's local time. It can be up to 60 minutes before dusk; Astryx uses it as you typed it and warns you it's before dusk. Later times, such as 01:00, mean that time after midnight. A time more than 60 minutes before dusk or after dawn can't be used, so Astryx says so and starts at dusk.

Sending Targets from a Filter

Filter Targets is grayed out and reads "Filter Targets (none loaded)" until you send it a list. To send one:

1. In **Target Selection**, set the filters (see Finding Targets).
2. In the Results card, click **Send to Optimizer**. Astryx sends every result, not just the ones on screen, and says how many it sent.
3. Open **Target Optimizer**.
4. Under **Source**, choose **Filter Targets**. It now shows the count, for example "Filter Targets (240 targets)".

The list stays until you send another one or close Astryx.

The Results

The line at the top of the **Suggested Targets** card says how many targets are shown out of how many were checked, and why the rest were left out:

Reason	What it means
never rises above minimum altitude	The target stays below your Min Altitude all night.
window too short	The target is above your Min Altitude for less than an hour.
below top 37	The target qualified, but only the 37 best are shown.

A Result Card

The cards are ranked by **Composite Score**, best first. Each card has:

- The target's designation, common name, type, and **Min/Max Size**, in arcminutes.
- **Window**: when the target is above your Min Altitude during the night, and for how many hours.
- **Peak**: the highest it gets during the window, and when.
- **Moon**: how close the Moon comes to the target, in degrees, while the Moon is up during the window. If the Moon is down all through the window, it's the distance at the middle of the window.
- **Dips below min altitude**, with the times, if the target drops below your Min Altitude and comes back up during the night. The Window covers only one of the two parts.
- Four scores from 0 to 100, **Window**, **Altitude**, **Centering**, and **Moon**, and the **Composite Score** that combines them.

Score	What scores high
Window	A long window. One hour scores 0; eight hours or more scores 100.
Altitude	A high peak. Peaking at your Min Altitude scores 0; straight overhead scores 100.
Centering	Crossing the meridian, its highest point, in the middle of the night.
Moon	The Moon down, faint, or far away during the window.

The Composite Score weighs Window and Moon at 30% each, Altitude at 25%, and Centering at 15%. Target Optimizer Scoring gives the formulas.

> **Note**
> The Optimizer uses your Min Altitude and the location's horizon profile, so a target behind your trees ranks lower or is left out. To see how much the horizon costs a target, open it in **Daily Visibility** and turn **Horizon** off and on.

What You Can Do with a Card

Button	What it does
Daily Visibility	Makes the target the Current Target and opens Daily Visibility for the Optimizer's date, with your horizon profile on.
Pin	Adds the target to Pinned Targets, ready for the Sequence Planner.
×	Removes the card from the results. Use it to drop targets you don't want before looking at Best Combinations.

Best Combinations

The Sequence Planner schedules targets one after another, so a long night can hold two or three of them. Click **Best Combinations** at the top of the results to see which ones share the night well. Click **Individual Targets** to go back to the cards.

The combinations are built from the targets in the results, so anything you removed with × is left out. The combinations come in three lists, of up to five each: **Solo Targets**, **Pairs**, and **Triplets**. Each combination shows:

- **Coverage**: the hours from the earliest window's start to the latest window's end, out of the hours in the night.
- **Combo score**: how good the combination is. It counts both the targets' Composite Scores and the hours each one can have to itself, so it isn't a 0-to-100 score; use it to compare combinations with each other.
- Each target's **Window**, **Peak**, and **Score**.

When two targets are up at the same time, they have to share those hours, so a pair that follows one another across the night scores better than one that's up at the same time.

Click **Replace Pinned Targets** to unpin all your Pinned Targets and pin the combination's targets instead. Astryx asks first.

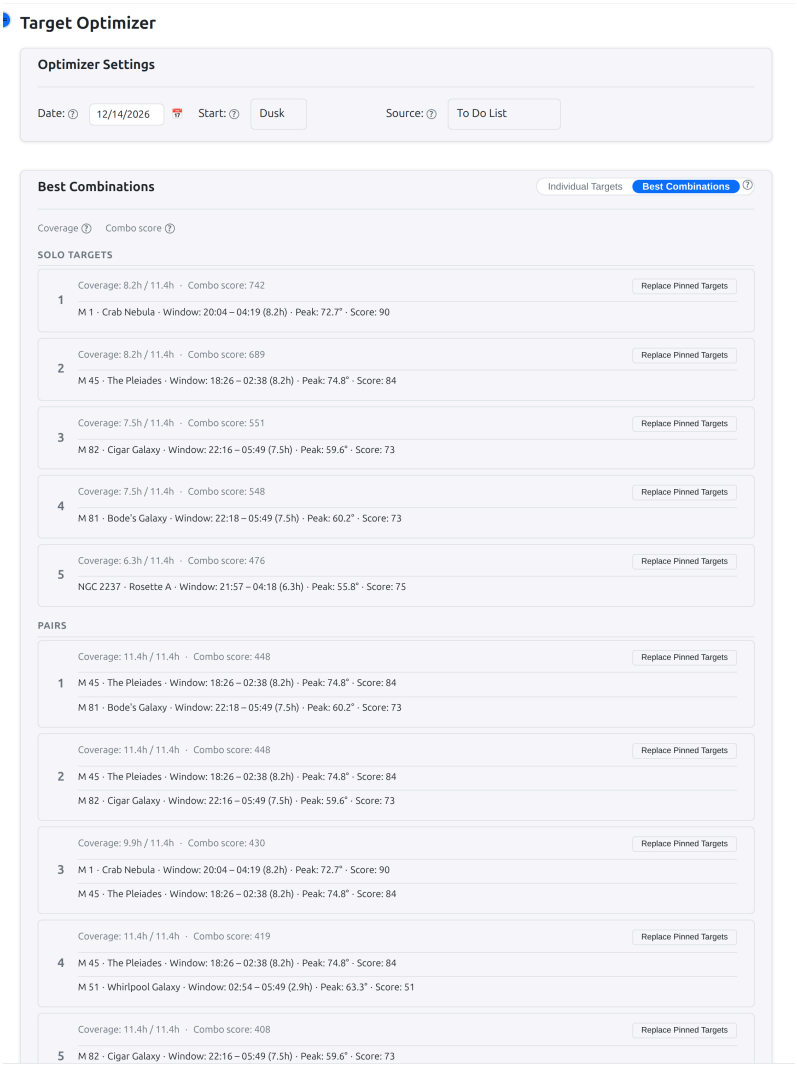


Figure 18: Best Combinations: the Solo Targets, Pairs, and Triplets lists, with Coverage, Combo score, and Replace Pinned Targets on each card.

> **Why this matters**
> The best single target isn't always part of the best night. A target that is up from dusk to dawn uses the whole night by itself, while two targets that rise and set at different times can each get a good run at their best. Best Combinations looks at the night as a whole.

Coming Back to the Results

The results stay while Astryx is open, so you can look at a target in Daily Visibility or the Viewfinder and come back to them. The **Date**, **Start**, and **Source** come back too, as they were when you ran the Optimizer, and the card's **Daily Visibility** button always opens the night the results are for. Change any of them for new results.

If there's nothing to rank, for example when your To Do List is empty or no target clears your Min Altitude that night, the results area says why.

Tips

> **Astryx Tip**
> Run the Optimizer on your To Do List for the next few nights in turn. A target that ranks low tonight because of the Moon may top the list next week.

> **Astryx Tip**
> If you can only stay up part of the night, set **Start** to **Custom** with the time you'll be ready. Targets that set early drop down the list.

Related

- Your First Night: the Optimizer as part of planning a night.
- Finding Targets: building a list to send.
- To Do List: keeping a shortlist to rank.
- Daily Visibility: checking a result in detail.
- Target Optimizer Scoring: the formulas.

Chapter 10

Sequence Planner

> **At a glance**
> The **Sequence Planner** turns your Pinned Targets into a plan for one night: which target to image when, how many exposures each gets, and when autofocus, guide calibration, and meridian flips will interrupt. It builds the best plan it can as soon as it opens; you can then adjust it.

What You'll See

Open **Sequence Planner** in the sidebar. It plans the targets under **Pinned Targets** (see Introduction); with none pinned, Astryx says "No pinned targets. Please pin targets first." When no plan can be made, because nothing is pinned, the night has no astronomical darkness, or no pinned target is high enough for long enough, the planner clears the old plan and Target Allocation reads "No plan". From top to bottom, the view has four cards:

- **Session Settings**: the night, the place, and how long your rig takes for autofocus, meridian flips, guide calibration, and dithering. Click the card's title to fold it away once it's set.
- **Target Allocation**: one row for each target, with its exposure length and its share of the night.
- **Timeline**: the night as a bar, with each target's turn and every interruption.
- **Imaging Plan**: when to start and stop each target, and how many exposures you'll get.

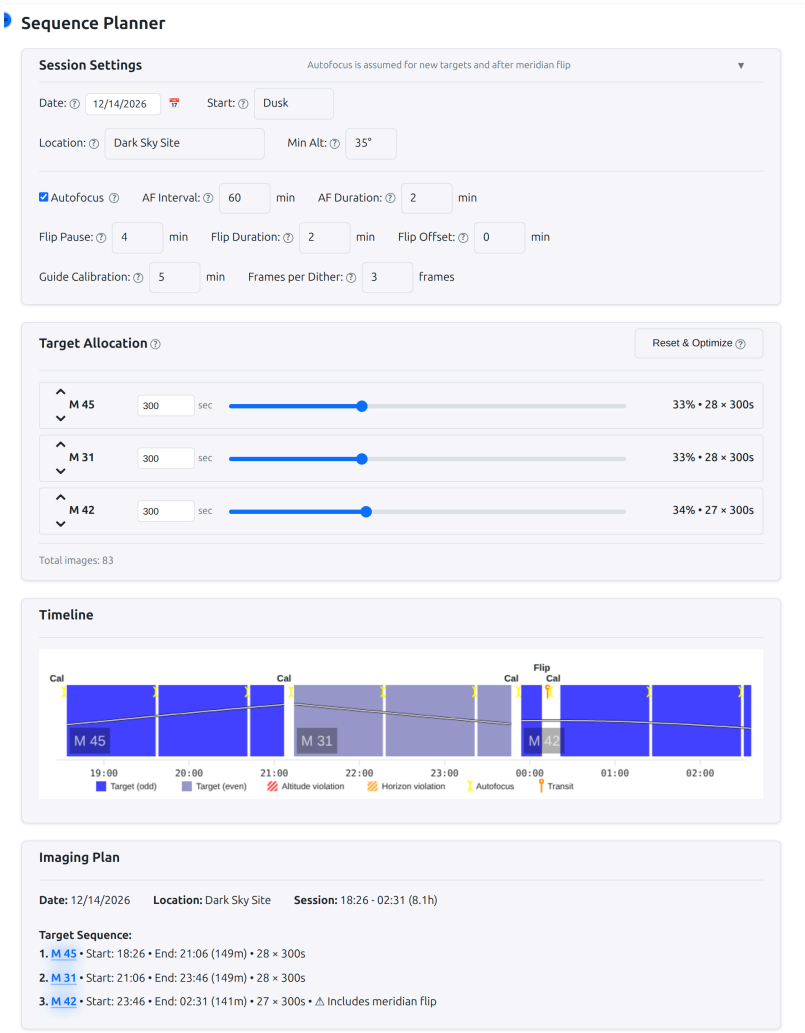


Figure 19: The Sequence Planner with three pinned targets: Session Settings, Target Allocation with three rows, the Timeline, and the Imaging Plan.

Session Settings

The Night

Setting	What it does
Date	The night to plan: the evening of this date and the morning after. It starts at today.
Start	**Dusk** starts at astronomical dusk. **Custom** shows a field for your own start, filled in with dusk; type the time as 24-hour HH:MM (2300 works too), in the location's local time.
Location	Where you'll image. It starts at the location chosen in the sidebar, and can be set to any of your locations.
Min Alt	The lowest altitude you'll image at. It starts at the Min Altitude in Settings; a different value is highlighted. Targets must also be above the location's horizon profile.

A Custom time can be up to 60 minutes before dusk; Astryx uses it as you typed it and warns you it's before dusk. Later times, such as 01:30, mean that time after midnight. A time more than 60 minutes before dusk or after dawn can't be used, so Astryx says so and starts at dusk.

The plan never starts before the first target is above your minimum altitude, and ends when the last target sets or at astronomical dawn, whichever comes first.

Overheads

The note in the card's title, "Autofocus is assumed for new targets and after meridian flip", sums up the autofocus rule. The rest of the card describes your rig. Astryx remembers these settings, so you set them once.

Setting	What it does
Autofocus	Untick it if you don't autofocus.
AF Interval	Minutes between autofocus runs on the same target: 0, 30, 60, 90, or 120. With 0, autofocus runs only when a target starts and after a meridian flip.
AF Duration	How long an autofocus run takes, in minutes.
Flip Pause	How long imaging stops before the meridian flip, in minutes.
Flip Duration	How long the flip itself takes, in minutes.
Flip Offset	How many minutes after the target crosses the meridian the flip starts. Match your capture software.
Guide Calibration	How long a guide calibration takes, in minutes. It runs when each target starts and after each flip.
Frames per Dither	How many exposures between dithers. Choose 0 if you don't dither; the label then reads "no dither".

> **Why this matters**
> Overheads add up. With a calibration and an autofocus at the start of each target, an autofocus every 30 minutes, and a meridian flip, a three-target night can lose most of an hour. A plan that ignores them promises exposures you won't get.

> **Note**
> Before the Flip Pause begins, your capture software checks whether there's time for another complete exposure. If there isn't, it stops and waits. So every meridian flip costs an extra, random amount of time, anywhere from nothing to one exposure length. Because it's random, Astryx can't plan for it, and a target that flips may end up an exposure or so short.

Time Between Exposures

Your camera also needs a few seconds between exposures to save each one, and a dither takes time to settle. There's no setting for these: Astryx starts with 5 seconds between exposures and 25 seconds for each dither, and improves both figures each time you analyze a session log in Log Analysis. The more logs you analyze, the closer the plan's exposure counts get to what your rig really does.

Target Allocation

Each row has:

- the up and down arrows at the left, to move the target earlier or later in the night,
- the target's name,
- the exposure length in seconds; it starts at 300,
- a slider for the target's share of the night, and
- the result, for example "40% 36 × 300s": the share and the number of exposures.

Total images below the rows adds up the exposures.

Moving a slider takes time from, or gives time to, the targets after it, split evenly among them; the targets before it keep their share. The last target's slider can't go above the share it had when the plan was built, because the night ends when that target sets, or at dawn.

Reset & Optimize throws away your changes to the order and the sliders and builds the best plan again. Your exposure lengths stay.

> **Note**
> Changing your Pinned Targets, or leaving the Sequence Planner and coming back, builds a new plan, and the exposure lengths go back to 300 seconds. Every other change keeps them.

When the Plan Changes

You change	What happens
Date, Start, Location, Min Alt, or the Pinned Targets	A new plan is built from the start.
Any other session setting	The exposure counts are worked out again; the order and shares stay.
An exposure length or a slider	The same.
The order, with the arrows	The same, and the night's start and end move to suit the new first and last targets.

The plan updates a second after your last change.

The Timeline

The timeline runs from the start of the plan to its end, with the hours labeled below it. Each target's turn is a blue block with its name; neighboring targets are in two shades of blue so you can tell where one ends and the next begins. Inside each block:

- The white line is the target's altitude, from 0° at the bottom of the block to 90° at the top.
- Yellow ticks at the top mark autofocus runs.
- An orange mark with a small circle on top marks the target crossing the meridian.
- **Cal** above the block marks a guide calibration, and **Flip** above **Cal** marks a meridian flip.
- Red crosshatching marks time when the target is below your minimum altitude.
- Orange hatching marks time when the target is behind your horizon profile.

The legend below the timeline names each of these.

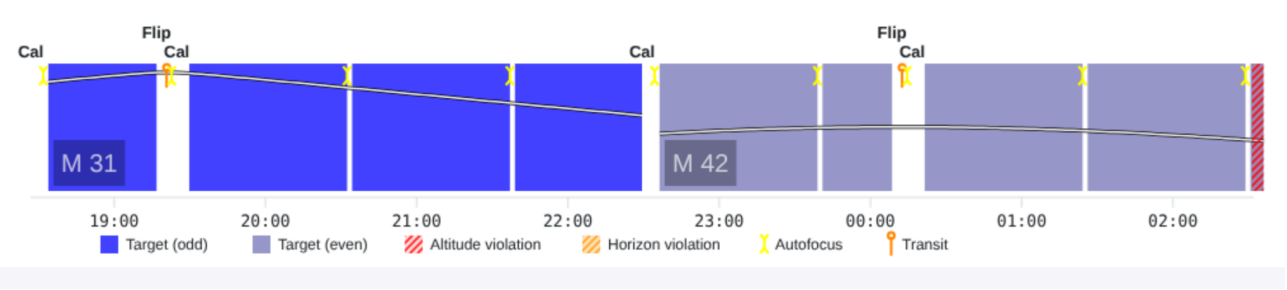


Figure 20: The Timeline for a two-target night: two shades of blue, the altitude lines, autofocus ticks, a transit mark, and Cal and Flip labels.

The Imaging Plan

The first line gives the **Date**, the **Location**, and the **Session**: the start and end times and the hours between them.

Target Sequence lists the targets in order. Each line gives the start and end times, the minutes left for imaging once the overheads are taken out, and the number and length of the exposures, for example “28 × 300s”. “Includes meridian flip” marks a target that flips during its turn.

Red lines under a target warn about time it can’t be imaged:

- **Altitude constraint**: the target is below your minimum altitude at the start or the end of its turn.
- **Horizon constraint**: the target is behind your horizon profile.

Each comes with its start and end times and its length. Move the handover with the sliders, or change the order, to shrink or remove them.

With four or more targets, a target that can’t get enough time is left out and listed at the bottom: “Not planned, not above the minimum altitude long enough tonight”.

Session Detail

Click a target’s name in the Target Sequence to see its turn event by event: each **Calibration**, **Autofocus**, **Imaging**, and **Meridian Flip**, with its start, end, and length. Click **Download PDF** to save the list for the night.

How the Plan Is Built

The planner tries to give every target at least two hours of usable imaging, since that’s about what you need for dithering to pay off. Then it gets as much imaging as possible out of the night, and then shares it as evenly as it can. Usable imaging is time spent exposing while the target is above your minimum altitude and horizon.

With two or three targets, it tries every order, and for each one, where to put the handovers and whether to avoid a meridian flip by handing over just before it. With four or more, up to a Messier marathon, it images whichever target will be lost soonest next. Sequence Planner Rules has the details.

The order you pinned the targets in doesn’t matter. The planner doesn’t take the Moon into account; check that with the Target Optimizer or Daily Visibility before you pin.

Tips

> **Astryx Tip**
> Start from the Target Optimizer. Its **Best Combinations** suggests targets that share a night well, and **Replace Pinned Targets** hands them straight to the Sequence Planner.

> **Astryx Tip**
> If a target only gets a sliver of the night, unpin it. Two targets with three hours each usually give a better result than three with two.

> **Astryx Tip**
> Copy the Imaging Plan into your capture software's sequence: the start times, the exposure lengths, and the counts. Astryx doesn't control your equipment.

Related

- Your First Night: planning a night from start to finish.
- Target Optimizer: choosing targets that share a night well.
- Getting Started: locations, horizon profiles, and the Min Altitude setting.
- Sequence Planner Rules: how the plan is built, in detail.

Chapter 11

Imaging Log

> **At a glance**
> The **Imaging Log** records what you've imaged. A **project** is one image you're working toward, such as a single target or a mosaic, and each night you spend on it is a **session**. **Programs** track a list of targets you want to work through, such as the Messier catalog. **Reports** and **Activity** show your progress.

What You'll See

Open **Imaging Log** in the sidebar. It has four tabs:

Tab	What it holds
Projects	Your projects, each with its sessions.
Programs	Lists of targets to work through, with your progress on each.
Reports	Summaries: which catalogs you've covered, progress on each program, and your projects by status.
Activity	A calendar of the nights you imaged.

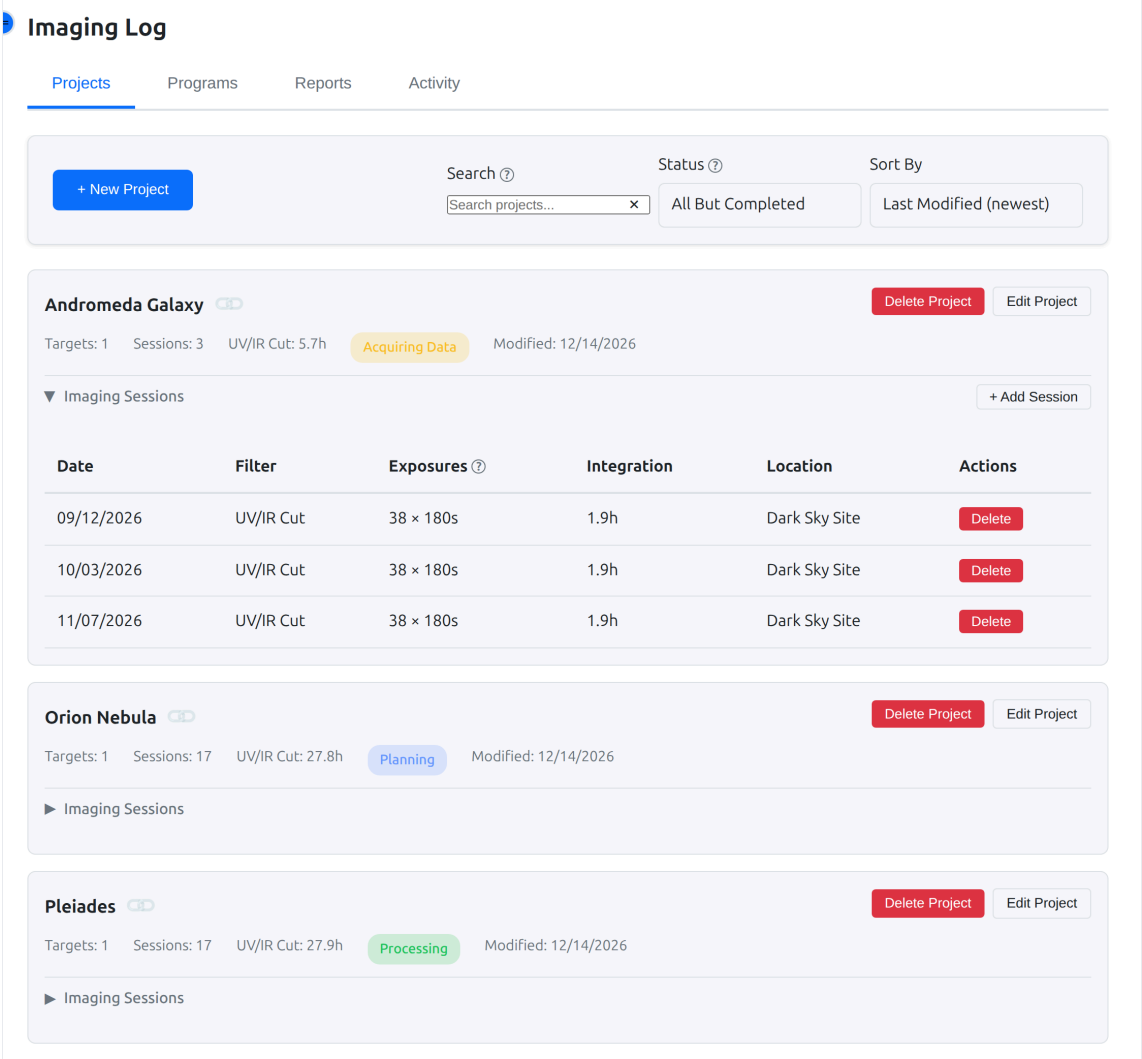


Figure 21: The Projects tab with three project cards, one of them opened to show its sessions table.

Projects

Creating a Project

1. On the **Projects** tab, click **+ New Project**.
2. Type a **Project Name**, for example “Horsehead Nebula Region”.
3. Under **Targets**, type part of a designation or common name, and click the target in the list. Add as many as the image covers; a mosaic or a field with several objects can have more than one.
4. Leave **Status** at **Planning**, or choose where the project stands.
5. Click **Save Project**.

A target’s other designations appear next to it, for example “(also: NGC 1976)”. **Remove** takes a target out of the project.

The project form also has:

- **Published Image Link**: a web address for the finished image, for example on AstroBin. It must start with http:// or https://.
- **Notes**: anything you want to remember about the project. Click **Expand** for a larger box.

The Project Card

Each project is a card. Its lines show:

- the name, with a link to the published image if you added one,
- **Targets** and **Sessions**: how many of each,
- the integration so far, by filter, for example "L: 6.5h Ha: 3.0h",
- the **Status**, and the date the project was last changed (**Modified**).

Edit Project opens the project form again. **Delete Project** deletes the project and all its sessions, after asking.

Click the card to open its **Imaging Sessions**: a table of the sessions, oldest first, with each one's **Date**, **Filter**, **Exposures**, **Integration**, and **Location**. Click the card again to close it.

Status

A project moves through five steps:

Status	Meaning
Planning	You haven't started imaging.
Acquiring Data	You're collecting exposures.
Acquisition Complete	You have all the data you want.
Processing	You're working on the image.
Completed	The image is finished.

Only **Completed** projects count as imaged in Programs and Reports.

Finding a Project

- **Search** finds projects by name or by target designation. It searches all projects, whatever the Status filter says.
- **Status** shows projects at one status. It starts at **All But Completed**, so finished projects are out of the way; choose **All Statuses** to see everything.
- **Sort By** orders the list by **Last Modified** (newest or oldest), **Name** (A-Z or Z-A), or **Status**.

Sessions

Adding a Session

1. Click a project to open it, then click **+ Add Session**.
2. Fill in the **Equipment** section: **Date**, **Location**, **Telescope**, **Sensor**, and **Filter**, and if you like, the camera's settings.
3. Click **Calculate Moon Data** to fill in the Moon.
4. Set the night's **Clouds**, **Smoke**, **Seeing**, and **Transparency**.
5. Under **Acquisition**, enter the **Sub Length (seconds)** and **Original Exposures**.

6. Click **Save Session**.

A new session copies the location, telescope, sensor, filter, and sub length from the project's most recent session, so a second night on the same target takes only a few changes.

To change a session later, click its row in the project's table. To delete it, click **Delete** in its row.

> **Note**
> A session needs a **Date**, a **Location**, a **Telescope**, a **Sensor**, a **Filter**, a **Sub Length**, and at least one **Original Exposure** before it can be saved, and its date can't be in the future. If you record the session before you start imaging, enter the number of exposures you plan, for example from the Sequence Planner, and correct it in the morning.

The Session Fields

Section	Fields
Equipment	**Date**, **Location**, **Telescope**, **Sensor**, **Filter**, plus **Rotation (°)** (the camera angle your plate solver reported), **Temp Setpoint (°C)**, **Bin**, **Gain**, and **Offset**.
Moon & Conditions	**Moon Set**, **Moon Rise**, **Illumination (%)**, **Angle from Moon (°)**, and the four condition lists.
Acquisition	**Sub Length (seconds)**, **Original Exposures**, **Used**, and **Integration Time**, which Astryx works out.
Notes	Anything about the night. Notes are kept as plain text; if you write them in Markdown, they render when you paste them into a Markdown document.

The equipment lists hold what you saved in Admin Tools (see Getting Started). The condition lists start at **Unknown**:

List	Choices
Clouds	None, Intermittent, Hazy
Smoke	None, Light, Moderate
Seeing	Poor, Below Average, Average, Good, Excellent
Transparency	Poor, Below Average, Average, Good, Excellent

The Moon

Calculate Moon Data needs the **Date** and **Location**. It uses the project's first target, and fills in:

- **Moon Rise** and **Moon Set**, in the location's local time, between noon on the date and noon the next day. When only one of them falls in that window, the other is the rise just before it or the set just after it, so both are filled in,
- **Illumination (%)**, the Moon's illuminated fraction over the night, and
- **Angle from Moon (°)**, the closest the Moon comes to the target while the target is above your Min Altitude.

Exposures and Integration

Original Exposures is how many exposures you took. **Used** is how many you kept after throwing out the bad ones: trailed stars, clouds, satellite trails you couldn't remove. It can't be more than Original Exposures.

Integration Time is Used times Sub Length, for example "115 × 300s = 9.6h". Only **Used** exposures count: the project card, the sessions table, and the integration by filter all show 0 for a session until you fill in Used.

> **Why this matters**
> Integration time, the total exposure you stack, is what decides how deep and smooth an image gets. Doubling it improves the signal-to-noise ratio by about 40%. The log adds it up for each filter, so you can see which filter needs another night.

Programs

A program is a list of targets you want to image over months or years. Its progress counts the targets that are in a **Completed** project, under any of their designations: a Completed project on NGC 1976 counts toward M 42 in a Messier program.

Creating a Program

Click **New Program**, type a **Program Name**, and choose a **Program Type**:

- **Catalog Pattern** covers a numbered catalog: choose a **Catalog Prefix** and type the **Maximum Number**. For example, M and 110 make the Messier catalog. Targets that don't exist in the database still count toward the total. Each number counts once: NGC 5194 and NGC 5194A are the same entry, and a number above the Maximum Number doesn't count, so a program never passes 100%.
- **Manual List** takes a **Target List**, one designation per line. Astryx matches each line against the database, including other designations, and shows which lines **Matched** and which **Failed**. **Copy Missing Targets as CSV** copies the failed lines so you can fix them.

Click **Save Program**. To make a program from a filter in Target Selection instead, use **Create Imaging Program** (see Finding Targets).

The Program Card

Each program shows its **Targets**, how many are **Imaged** and what percentage that is, how many are **Remaining**, a status (**Not Started**, **Started**, or **Complete**), and a progress bar. **Edit Program** and **Delete Program** change or remove it.

> **Note**
> Adding a target to a project warns you if its number is above a Catalog Pattern program's Maximum Number, for example NGC 7841 in a program that stops at 7840.

Reports

The **Reports** tab has three cards:

Report	What it shows
Catalog Coverage	For each catalog, how many targets you've imaged, counting each target's other designations too.
Program Progress	Each program's progress, with **Targets Completed** (click a project's name to go to it) and, for a Manual List, **Targets yet to be imaged**.
Project Status	For each status, how many projects, targets, and sessions.

Catalog Coverage and Program Progress count only **Completed** projects.

Activity

The **Activity** tab is a calendar of one year, a square for each day, like the contribution calendar on GitHub. A day you imaged is colored; the brighter the color, the more integration you collected that night, compared with your other nights that year. Like the rest of the Imaging Log, it counts **Used** exposures. A session whose subs haven't been processed yet (**Used** is 0) counts the subs you took instead, until you enter Used, and its night is marked with a dot; the legend calls it **Not processed yet (shaded by subs taken)**. Hold the pointer over a day to see each session's used and taken subs. Drag the slider above the calendar to choose the year.

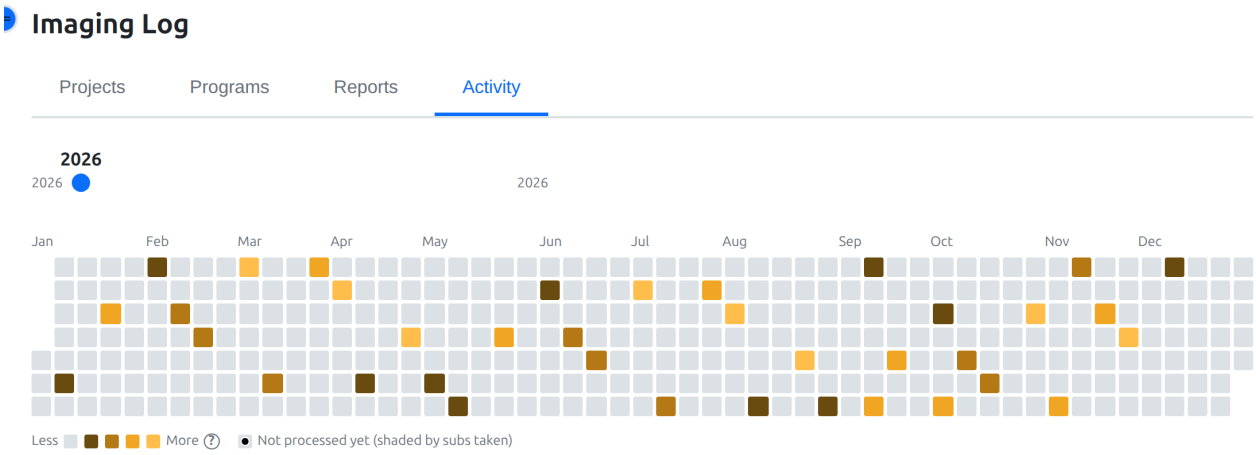


Figure 22: The Activity tab for a year with several dozen imaging nights.

Tips

> **Astryx Tip**
> Create a project when you start planning a target, not after the first night. Its Planning status reminds you it's in the queue, and the first session copies nothing until there is one.

> **Astryx Tip**
> Set a project to **Completed** when the image is finished. That's what moves your programs and Catalog Coverage forward.

Related

- Your First Night: recording a night as part of the routine.
- Finding Targets: making a program from a filter.
- Getting Started: locations, telescopes, sensors, and filters.

Chapter 12

Log Analysis

> **At a glance**
> **Log Analysis** reads the logs your ASIAir and PHD2 write during a night, and tells you what happened: how many subs were clean, where the time went, what went wrong and when, and what to change. Each log you analyze also teaches the Sequence Planner how long your rig really takes between exposures.

What You'll See

Open **Log Analysis** in the sidebar. The **Session Log Analysis** card has one drop box and a **Browse...** button for the night's two logs:

- **Session Log**: the ASIAir's Autorun log for the night, a .txt file.
- **PHD2 Guide Log**: the PHD2 guide log for the same night, also a .txt file, with a name like PHD2_GuideLog_2026-10-07_201500.txt.

Astryx tells the two apart by their contents. The name of each loaded log appears beside the button.

Below are **Guide Stability** and **Settle Time**: your ASIAir's settle settings (after a dither, guiding must stay under this many arcseconds for this many seconds). Neither log records them, so enter them here. Astryx remembers them, so you only change them when you change them on the ASIAir.

Loading the Session Log adds the **Combined Report** below them, titled Combined Report — *targets* — *date*. Click its title to open or close it. The PHD2 Guide Log adds the guiding analysis; without it, the report leaves out every guide figure and says so. A PHD2 Guide Log on its own shows only a note asking for the Session Log.

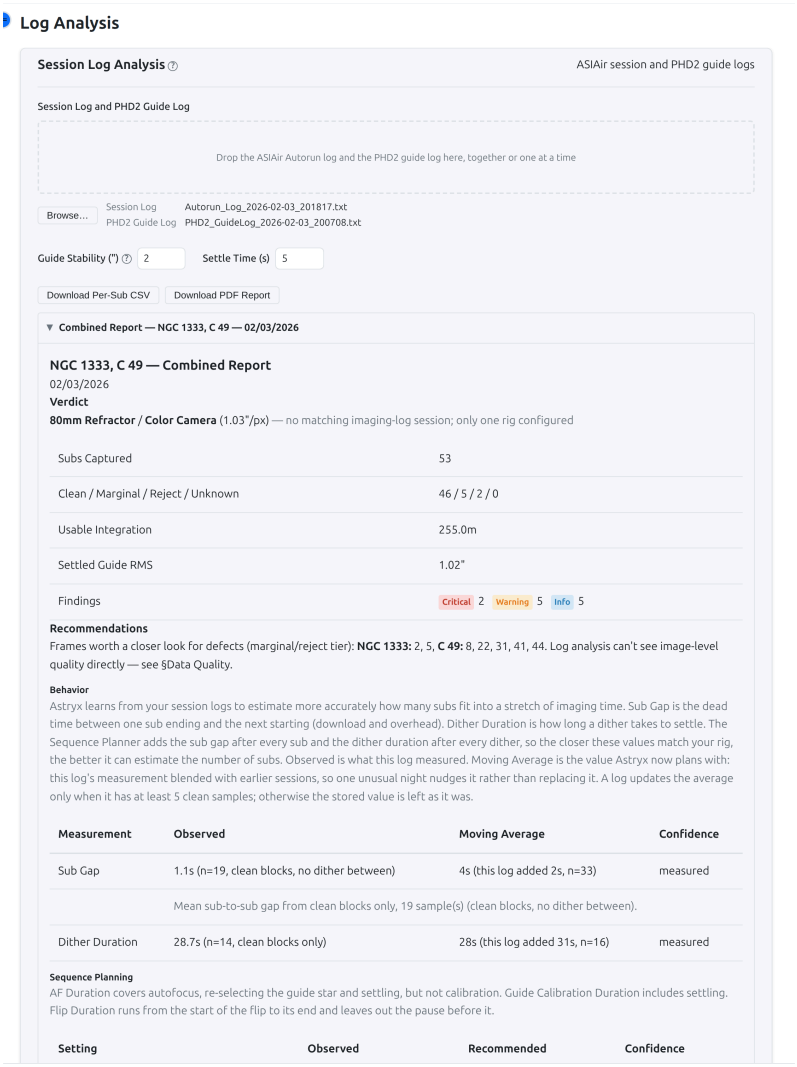


Figure 23: Log Analysis with both logs loaded: the drop box, the two file names, and the Combined Report open at its Verdict.

Loading a Night

1. Copy the night's logs from your ASIAir to your computer.
2. Drag both logs onto the box, or click **Browse…** and select them.

You can load them together or one at a time, in either order. Loading another log of the same kind replaces the one loaded, and rebuilds the Combined Report. If the guide log doesn't overlap the Session Log in time, Astryx warns you that they may be from different nights.

If you recorded the night in the Imaging Log (a session with the same date), the Combined Report uses that session's telescope, sensor, and location. Without one, it uses your telescope and sensor if you have only one of each, and leaves out what needs a location.

A night with only flats, darks, or bias frames has no light frames to analyze; the Combined Report says so and stops there.

What the Logs Teach the Sequence Planner

Each time you load a Session Log, Astryx measures two things the Sequence Planner needs:

- the **sub gap**: how long the camera takes between the end of one exposure and the start of the next, and
- the **dither duration**: how long a dither takes to settle.

A gap that isn't the camera's own overhead is left out of the sub gap: one where the ASIAir re-targets between subs, or a pause more than a couple of seconds longer than the night's typical gap. One long pause would otherwise make every later plan count too few subs.

Each stored figure is the average of your 20 most recent nights, to a tenth of a second, so a single unusual night doesn't throw it off. Most recent means by the night's date, not by when you analyzed it, so the order you load logs in doesn't matter, and an older night than those 20 doesn't change the figures. A night with too few clean samples doesn't count toward a figure. Until you've analyzed 20 nights, the figures are the average of those you have; the very first replaces the starting values.

The **Behavior** table in the Combined Report's Recommendations shows each night's measurement and the stored figure. After the stored figure it says what it was before this log, for example `1.1s (was 1.2s; this log's 0.9s included, average of the last 20 nights)`, or why this log didn't change it.

> **Note**
> Each night's Session Log updates the stored figures once. Astryx remembers which logs it has used, by their start times, so loading the same log again leaves the figures alone and tells you so. The report itself appears as usual. In the desktop app, The Command Line can start the figures over, and can analyze many nights' logs in one go. Start them over after changing something that alters them, such as your camera, mount, dither size, or settle settings, so old nights don't hold them back.

The Combined Report

Verdict

The headline numbers: **Subs Captured**; how many were **Clean / Marginal / Reject / Unknown**; **Usable Integration**; **Settled Guide RMS**, the guiding error once each dither had settled; and the **Findings**, counted by severity: **Critical**, **Warning**, and **Info**. The telescope and sensor appear above them when Astryx knows them.

Every sub is given a tier:

Tier	Meaning
Clean	Nothing flagged.
Marginal	Usable, but something was off: guiding worse than usual for the night, a start before the dither settled, or a guide failure during the exposure.
Reject	Guiding much worse than usual for the night, or the exposure was cut short.
Unknown	No guide data for the exposure, so no tier could be given.

"Usual for the night" means compared with the night's own median guiding, not a fixed number. Combined Report Details gives the limits.

Recommendations

First, the frames worth a closer look: every Marginal and Reject sub, by image number. Then up to three tables:

Table	What it covers
Behavior	The sub gap and dither duration: tonight's measurement and the running average the Sequence Planner uses.
Sequence Planning	AF Duration, Guide Calibration Duration, and Flip Duration, measured tonight, with the value to set in the Sequence Planner. A note says what each duration includes.
Guiding Configuration	PHD2 settings: Search Region, Star Mass Tolerance, Guide Star Selection, and Calibration Timing. Star Mass Tolerance is flagged when PHD2 lost the guide star to a change in its brightness in more than one guide session.

Each row gives what was **Observed**, what's **Recommended**, and the **Confidence**. A row that needs a change says **Change suggested** under the setting's name, with the evidence below it. A setting with nothing to change is still listed, so you know it was checked.

The confidence tells you how far to trust a row:

Confidence	Meaning
Measured	Read straight from the log.
Derived	Calculated from the log and checked another way.
Inferred	A pattern that matches known behavior, not a direct measurement.
Copied	Your setting, shown for reference; nothing in the logs could check it.

Guiding Settings

Each guide setting, with what tonight's logs showed and what that means. It opens with a short summary: whether RA and Dec errors were balanced, whether the guide camera's resolution is the real limit, and which settings are worth a look.

Setting	What tonight showed	What the assessment tells you
Calibration Step	RA steps in each calibration, and its Dec	PHD2 aims for about 12. Far fewer gives a less accurate calibration; far more only takes longer. Extra steps are normal above Dec 60°.
Max RA / Dec Duration	Pulses that hit the limit	Many hits mean corrections are being cut short.
Corrected Trigger Accuracy	Frames that needed no correction, and the jitter from seeing	Whether small errors go uncorrected or guiding chases the seeing.

Setting	What tonight showed	What the assessment tells you
RA Aggression, Dec Aggression	Error persistence: how much each frame's error carries into the next	Errors that linger suggest corrections lag; errors that flip suggest they overshoot.
RA / Dec Balance	Settled RMS on each axis	RA much worse than Dec points to periodic error or RA balance; Dec much worse points to Dec backlash or balance.
Guide Stability	Dither settle times and timeouts	Timeouts mean your stability setting is too tight for the night's guiding.
Guide Exposure	—	Shown for reference.

A row worth changing says **Change suggested** under the setting's name, and says what to try. The aggression rows are marked inferred: try a suggestion for a few nights and compare the reports before keeping it.

Session Timeline

The night in order: imaging blocks, autofocus runs, guide calibrations, meridian flips, and anything flagged as a finding, with each one's **Time**, **Duration**, and, for imaging, **Guide Quality**. A row naming the target marks each change of target, and repeated identical events are shown once with a count.

A meridian flip takes two rows. **Meridian Flip — Waiting** is the time from imaging stopping to the flip starting: any wait because the next sub wouldn't have finished in time, then the Flip Pause and the Flip Offset. **Meridian Flip** is the flip itself.

Dithers, plate solves, and the routine start and stop of tracking are hidden. Tick **Show all events** to see them.

Summary

Where the night's time went: imaging, autofocus, guide calibration, meridian flip, and the rest, with each one's **Total Time** and **% of Session**. Time Astryx can't account for is listed, not hidden. Dither time is part of imaging time. Dithers that didn't settle are counted by how they ended, with the images they affected.

Per-Sub Frame Quality

One table for each target, with a row for each sub: **Image #**, **Start**, **RMS RA**, **RMS Dec**, **RMS Total**, **Peak**, **Temp**, and **Tier**. The Tier column also notes dropped guide frames and a start before the dither settled. Marginal and Reject rows are highlighted.

Guiding Analysis

The guide camera and mount, overall guiding statistics, and the results for each side of the pier (counting only guide sessions while imaging) and for dithering, with the **effective frames per dither**: how many subs your ASIAir took between dithers. Below them are the **Guide Sessions** table (each stretch of guiding between interruptions, with its RMS, **Peak RA**, **Peak Dec**, and SNR; short sessions and locks near the frame edge are noted below it) and the **Calibrations** table (each PHD2 calibration: **West Rate**, **North Rate**, **Orthogonality**, and any **Star Lost**).

Findings

Every problem the analysis found, most severe first, each starting with its level (**Critical**, **Warning** or **Info**) and followed by the evidence behind it. Flagged events in the Session Timeline start with their level the same way. An Inferred finding also says what other explanations it ruled out. Combined Report Details lists the checks.

Focus and Environment

The night's autofocus runs, with each one's **Trigger**, **Duration**, **Outcome**, **Temp**, and **Star Size**, and any trend in focus position with temperature, or in star size over the night.

Data Quality

How far to trust everything above:

- how many internal checks passed and failed,
- how many log lines Astryx didn't recognize,
- how many subs have no guide data,
- any settle scans that never ended in the log (left out of the durations rather than estimated),
- any meridian flip with no end line, for example when Autorun was stopped during the flip (the subs after it and the rest of the log are still read),
- **Meridian Flip Verification**, and
- **Stated Limits**: what log analysis can never see.

Meridian Flip Verification compares what your ASIAir actually did at the meridian flip with the **Flip Pause** and **Flip Offset** set in the Sequence Planner. A large **Delta** means the two have drifted apart; set the Sequence Planner to match your ASIAir. Your ASIAir also stops early when the next sub wouldn't finish before the Flip Pause; that time shows as its own row, **Wait for a sub**, and changes from flip to flip. The table needs the night's location, so it appears only when the night is in the Imaging Log; otherwise a note says what's missing. The log's times are read as the clock at that location, so the table comes out the same whatever time zone your computer is set to. Every time in the report is the time your ASIAir wrote in its log.

Saving the Report

- **Download PDF Report** saves the whole report as a PDF. Keep it with the night's subs: it's a lasting record of how each sub was taken, what went wrong, and which frames to check, long after the logs are gone. It also prints well. The timeline in it leaves out the hidden events, as the screen does with **Show all events** unticked, and each target's Per-Sub table starts on a new page.
- **Download Per-Sub CSV** saves the Per-Sub Frame Quality table for a spreadsheet.

What Log Analysis Can't See

Log analysis only knows what the ASIAir and PHD2 wrote down:

- Satellite and aircraft trails are invisible: the guide camera looks at a different patch of sky from the main camera.
- Image quality itself, such as gradients, focus, and haze within a sub, isn't in the logs. A Clean sub had nothing wrong in its guiding and timing; it may still not be a keeper.

- The brightness of the guide star isn't a reliable measure of cloud, since PHD2 loses a fading star and picks another rather than watching it fade.
- A check finds only what it was written to find. A new kind of problem shows up only as a failed internal check or unaccounted time in Data Quality.
- The limits behind the tiers and findings were tuned on one rig's logs. They're a good starting point, not universal.

Reading a Night

Start with the **Verdict** and **Recommendations**: they tell you quickly whether the night was clean and whether anything needs changing. If the Verdict mentions a mismatch, check **Data Quality** before trusting the numbers.

For a night that went wrong, tick **Show all events** in the Session Timeline and read it alongside **Findings**: together they usually show what happened and when. Then look at the frames the Recommendations list. The logs can point to likely problem frames; only looking at the images can confirm them.

Tips

> **Astryx Tip**
> Record the night in the Imaging Log before you analyze its logs. The report then knows your telescope, sensor, and location, and can check your meridian flip settings.

> **Astryx Tip**
> Analyze your logs regularly, not just after bad nights. Clean nights are what tune the Sequence Planner's exposure counts.

Related

- Sequence Planner: the settings the recommendations refer to.
- Imaging Log: recording the night, so the report knows your equipment and location.
- Combined Report Details: tiers, confidence, and every check.

Chapter 13

Utilities

> **At a glance**
> **Utilities** collects quick links and small tools: weather forecasts, satellite clouds, and a light pollution map for any of your locations, and a calculator that estimates how far a dust speck is from your sensor.

What You'll See

Open **Utilities** in the sidebar. It has two cards: **Sky Conditions** and **Dust Mote Distance Estimator**. Hold the pointer over a card's **?** for a reminder of what it does.

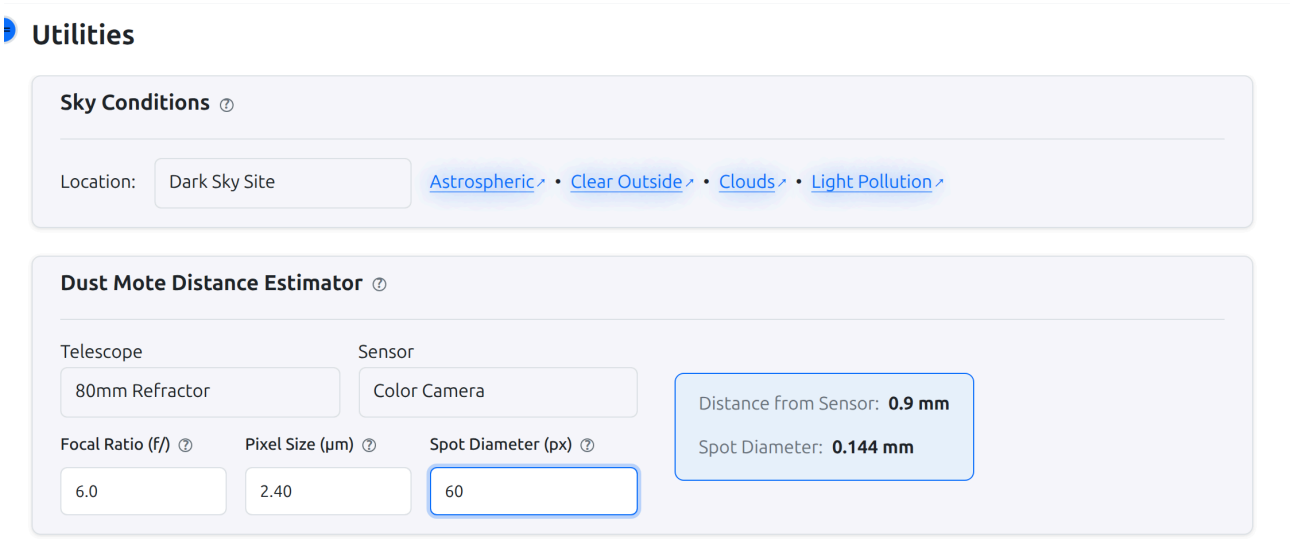


Figure 24: The Utilities view: Sky Conditions with its location and four links, and the Dust Mote Distance Estimator with a result.

Sky Conditions

Choose a **Location**; it starts on your current location. The four links beside it open at that location's coordinates, in your web browser, and need an internet connection. With no locations saved, the card says "No locations configured. Add locations in Admin Tools → Manage Locations."

Link	What it opens
Astrospheric	An astronomy forecast with cloud cover, transparency, and seeing, hour by hour.
Clear Outside	A forecast of low, medium, and high cloud, with the Moon and darkness.
Clouds	A live satellite map of the clouds around the location, from Zoom Earth.
Light Pollution	The Light Pollution Map, zoomed in on the location. Use it to see how dark your site is, or to find a darker one within driving distance.

> **Astryx Tip**
> Daily Visibility already shows cloud, wind, and dew for the next few days. Use these sites for a second opinion, and for seeing and transparency, which Daily Visibility doesn't forecast.

Dust Mote Distance Estimator

Dust on a filter or sensor window shows in your flats and lights as a soft disc, or a ring (a "dust donut") on a telescope with a central obstruction. The farther the speck is from the sensor, the larger the spot. Knowing the distance tells you which surface to clean.

1. In **Dust Mote Distance Estimator**, choose your **Telescope** and **Sensor**. Astryx fills in **Focal Ratio (f/)**, including the telescope's reducer or Barlow, and **Pixel Size (µm)**; you can also type them.
2. Measure the spot's diameter in pixels in your imaging software, and type it in **Spot Diameter (px)**. The edge is soft: measure across the spot where the edge has faded halfway.

The result is the estimated **Distance from Sensor**, in millimeters: the spot's diameter in millimeters times the focal ratio. The line below it, **Spot Diameter**, gives the spot's diameter in millimeters. If you imaged with binning, enter the binned pixel size, or measure on an unbinned frame.

The distance is approximate. Glass makes things look closer than they are, so dust seen through a filter or window is a little farther away than the estimate.

Compare the distance with your imaging train: a few millimeters usually means the sensor window, and farther away a filter or the rear lens of the flattener or reducer. Dust further up the train is too far out of focus to show as a spot.

Related

- Getting Started: adding locations, telescopes, and sensors.
- Daily Visibility: the forecast built into Astryx.
- Log Analysis: analyzing a night's logs.

Chapter 14

Backup and Restore

> **At a glance**
> Everything you enter in Astryx is stored only on your computer. A **Backup** saves it to a ZIP file you can keep somewhere safe; **Restore** brings it back, on the same computer or a new one. Back up regularly: a backup is the only copy that survives a failed disk, a cleared browser, or a lost laptop.

What a Backup Holds

A backup is one of two kinds:

Kind	What it holds	How often
User data	**Settings**, **Locations**, **Telescopes**, **Sensors**, **Filters**, **Pinned Targets**, your **To Do List**, **Imaging Projects / Sessions**, and **Imaging Programs**.	After every imaging night, or at least weekly.
Targets	The **Target/Best Months Database**: the ~14,600 targets and the Best Months you calculated for each location.	Once, and again after you calculate Best Months for a new location.

The target database is large and rarely changes, which is why it has its own backup.

Making a Backup

1. Open the system menu (≡) and click **Backup**.
2. Click **Select User Data** or **Select Targets**. User data is selected when the dialog opens.
3. Untick anything you want to leave out. The number after each item is how many records it holds.
4. Click **Save Backup**.

The dialog shows the **Filename**, and the backup's size before and after compression. The filename records the Astryx version, the database version, what the backup holds, and the date and time, for example Astryx-v1.11.1-d6-userdata-20261007-2103.zip.

Where the file goes:

- **Web version:** your browser's downloads folder.

- **Desktop app:** the **Backup Folder** from Settings, or, if you haven't set one, wherever you choose in the save dialog.

When it's done, **Data Saved** shows the file's name and size and how many records of each kind it holds.

> **Astryx Tip**
> Copy your backups somewhere other than the computer Astryx runs on: a cloud drive, a USB stick, or another computer. A backup on the same disk is lost with it.

Automatic Backups

With **Automatic backups on change** on in Settings, Astryx saves a user data backup by itself after you change your data: **Minutes before backup** after your last change, so a busy evening of edits makes one backup, not dozens. Changing the theme or the location in the sidebar doesn't count as a change.

- The wait is timed from your last change by the clock. If you close Astryx and open it again before the time is up, the countdown carries on where it left off. If the time ran out while Astryx was closed, the backup is made as soon as Astryx starts.
- Automatic backups go to your browser's downloads folder on the web version. On the desktop app they go to the **Backup Folder** from Settings, or to your Downloads folder if you haven't set one.
- On the web version, Astryx can't tell whether your browser actually saved the file. A browser can block a download it didn't expect, or ask where to save it and be dismissed. So the message says the backup was **downloaded**, not saved; check your downloads folder now and then. The desktop app writes the file itself and says if that fails.

See Getting Started for these settings.

The Backup Reminder

At the bottom of the sidebar, **Last backup** says how long ago you last backed up, or **never**. It turns amber after 7 days and orange after 14. When Astryx starts, and once a day while it's open, it also reminds you if your last backup is older than **Backup reminder every** in Settings.

Where your data lives depends on which Astryx you use. The web version stores it in your browser, where it can be lost if the browser's cache or site data is cleared, or by another browser operation such as an update. The desktop app stores it in a database on your computer, where it can be lost if the disk fails, or for any number of other reasons. Either way, a backup is what gets it back.

Click **Last backup** for **About Your Data Storage**, which explains why backups matter; its **Make a Backup** button opens the Backup dialog.

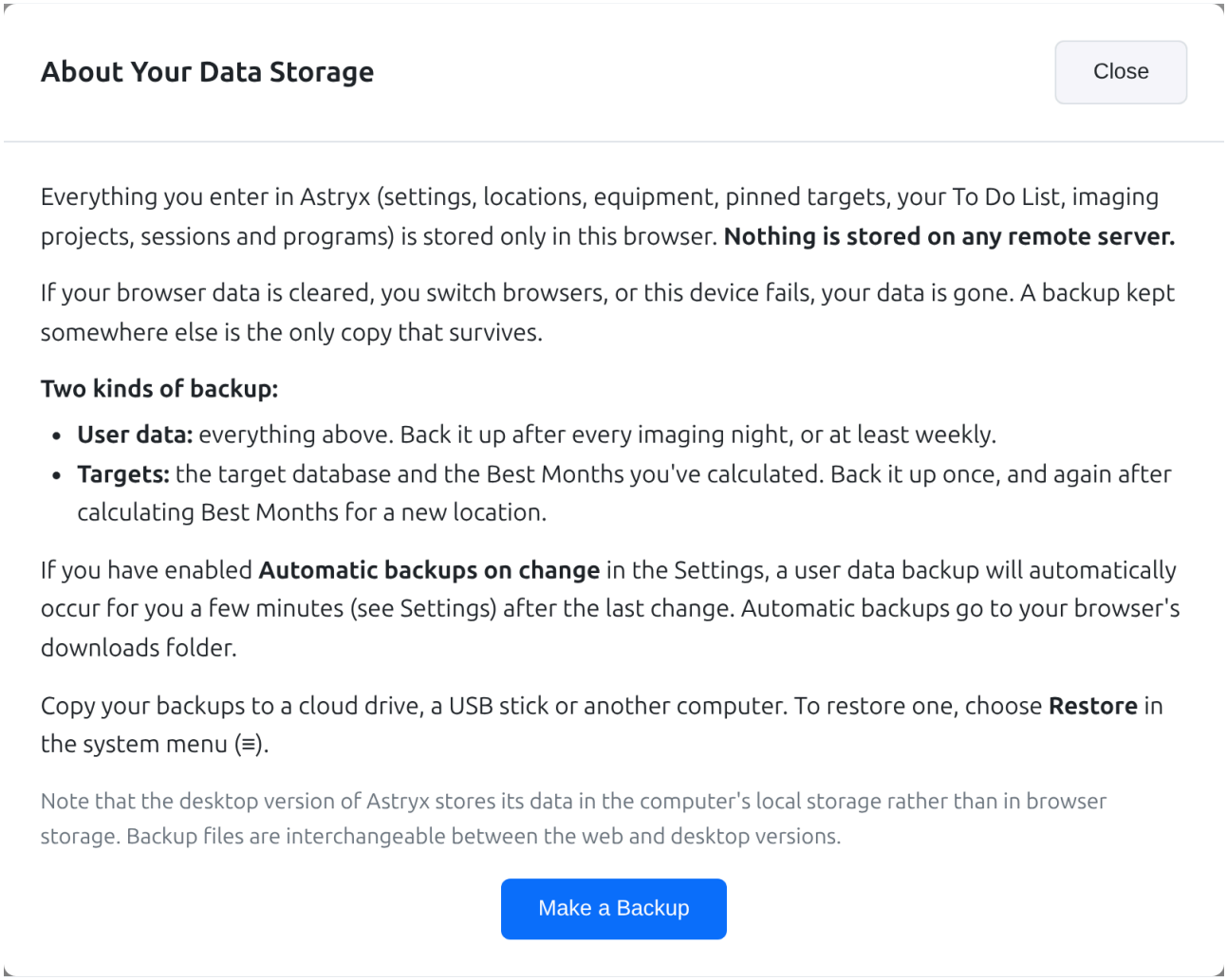


Figure 25: About Your Data Storage in the web version, opened from Last backup at the bottom of the sidebar.

Restoring a Backup

1. Open the system menu (≡) and click **Restore**.
2. Choose the backup file. On the web version, click **Browse Files**, then **Continue**.
3. Read the **Restore Backup?** dialog (below).
4. Untick anything you don't want to restore.
5. Tick the box that confirms you've backed up your current data.
6. Click **Restore**.

Astryx replaces each selected kind of data with the backup's copy, then reloads.

> **Note**
> Restoring replaces, it doesn't merge. A restored To Do List, for example, loses anything you added since the backup was made. If you're not sure, back up your current data first, so you can go back.

The Restore Backup? Dialog

The top of the dialog describes the file: when it was **Created**, the Astryx **Version** and **Database** version that made it, the **Target** database version, and its **Size**. Below that, Astryx compares the backup with your current data:

Message	What it means
Identical to current data — restore not needed	Nothing would change.
Backup data is newer than current data — restore recommended	The backup has changes your current data doesn't, for example from another computer.
Backup data is older than current data	Restoring would undo changes made since the backup. Backups over 30 days old are marked in yellow, and over 90 days in orange.

The last two show when each was last changed.

Each kind of data in the backup has its own line, with how many records you have now and how many the backup holds, for example "12 → 15", and whether it's **identical** or **differs**. Imaging projects and sessions are restored together.

A backup made by a version of Astryx with a different database version can't be restored; Astryx shows **Version Mismatch** instead.

Moving to a New Computer

1. On the old computer, make a user data backup and a targets backup.
2. Install Astryx on the new computer, or open the web version there.
3. Restore the targets backup, then the user data backup.

The same works between the web version and the desktop app.

Related

- Getting Started: the backup settings.
- Introduction: where Astryx keeps your data.

Chapter 15

Admin Tools

> **At a glance**
> **Admin Tools**, in the system menu, holds the jobs you do now and then: setting up locations and equipment, calculating Best Months, and keeping the target database up to date.

What You'll See

Open the system menu (≡) and point to **Admin Tools**. It has six items:

Item	What it does
Calculate Best Months	Works out each target's Best Month and Observable months for a location.
Manage Equipment	Adds, changes, and removes telescopes, sensors, and filters.
Manage Observer Locations	Adds, changes, and removes your locations and their horizon profiles.
Clear All Targets	Deletes the target database.
Check for Target Updates	Downloads a newer target database, if there is one.
Import Target Database	Not available at present; it's grayed out.

Getting Started covers **Manage Equipment** and **Manage Observer Locations**.

Calculate Best Months

Each target's **Best Month**, **Peak Altitude**, and **Observable** months depend on where you are, so Astryx works them out for each of your locations and stores them with the targets. The Visibility filter in Target Selection, the To Do List's grouping by Best Month, and the summary line in Yearly Observability all use them.

You rarely need to start this yourself. Astryx calculates Best Months on its own when you choose a location in the sidebar that doesn't have them yet, and when it starts; a message at the bottom of the screen shows the progress.

When you change **Min Altitude** in Settings, Astryx offers to calculate them again for the location in the sidebar, and for each other location when you choose it.

After Astryx's target database has been updated, it calculates them again on its own for every location that had them, one location after another.

Calculate them again yourself after you move a location's coordinates:

1. Open the system menu (≡), point to **Admin Tools**, and click **Calculate Best Months**.
2. Under **Location:**, choose a location, or **All Locations**.
3. Click **Calculate**.

A progress bar shows each target as it's done, with an estimate of the time left. Click **Cancel** to stop; a location's Best Months are saved only once it's finished.

When it's done, the dialog shows how many targets were checked, how many are **Visible targets** from the location, and how many are **Not meeting criteria**, that is, never well placed there.

Best Month Calculation explains how the months are chosen.

> **Why this matters**
> Best Months take time to calculate: every target, for every night of the year. Working them out once and storing them is what lets the Visibility filter search all ~14,600 targets instantly.

Clear All Targets

Deletes every target from the database, along with the Best Months. Your locations, equipment, Pinned Targets, To Do List, and Imaging Log are kept. Astryx asks first.

You'd only do this to start the target database afresh. Use **Check for Target Updates** afterwards to download it again.

Check for Target Updates

Astryx's target database is updated from time to time with corrections and new objects. **Check for Target Updates** compares your copy with the latest:

- "Target database is already up to date" means there's nothing to do.
- Otherwise Astryx downloads the new database and reloads. Best Months are then calculated again for every location that had them, starting with the one chosen in the sidebar. If the download or the save fails, your current target database is kept.

It needs an internet connection. The version of your target database is the number after "t" at the bottom of the sidebar, for example "v1.11.1 d6 t12".

Related

- Getting Started: locations, horizon profiles, and equipment.
- Best Month Calculation: how Best Months are chosen.
- The Target Database: where the targets come from.

Chapter 16

Recipes

> **At a glance**
> Each recipe here is a common job, done from start to finish across several views. The chapters for each view have the details.

Image Two Targets in One Night

Use the whole night by imaging one target while it's high in the evening, then switching to a second as the first sinks.

1. Add a handful of candidates to your **To Do List** (see To Do List).
2. Open **To Do List** and set **Sort by:** to **Rise Time**, with the chart showing. The chart's left edge is dusk and its right edge is dawn.
3. Choose the evening target: a bar that starts at the left edge and ends around the middle of the night. A target that crosses the meridian around dusk looks like this; it's at its highest as the night begins.
4. Choose the late target: a bar that starts before the first one ends and runs to the right edge.
5. Pin both: click each one in the To Do List and click **Pin** in its Target Details.
6. Open **Sequence Planner**. It images the evening target first, because it sets first, and hands over to the late target before the first sinks below your minimum altitude.
7. Check the **Timeline**. If the handover leaves one target with much less time than you want, move the slider in **Target Allocation**.

> **Astryx Tip**
> An evening target that has already crossed the meridian at dusk never needs a meridian flip. If the late target's turn starts after it crosses the meridian too, the whole night runs without a flip.

> **Astryx Tip**
> To let Astryx find the pair, run the Target Optimizer on your To Do List and click **Best Combinations**. Under **Pairs**, a pair whose windows follow each other across the night, with a high **Coverage**, is what you're after. **Replace Pinned Targets** pins it.

Image a Target Over Several Nights

Most targets need more integration than one night gives. To spread a target over several nights:

1. Open **Yearly Observability** for the target to see its season: the months where the line is high and the background dark.
2. In the **Imaging Log**, click **+ New Project** to create a project. Set its **Status** to **Acquiring Data** when you start.
3. Before each night, check it in **Daily Visibility**, and step through the nights with ▸ to find the next clear, dark one.
4. After each night, add a session with **+ Add Session**, and fill in **Used** once you've checked the subs.
5. Watch the integration by filter on the project card. When you have enough, set the status to **Acquisition Complete**.

Image a Whole Catalog

To work through the Messier catalog, or any numbered list:

1. In **Target Selection**, click **Reset**, tick only **Messier** in **Catalog**, and clear **Min Size (′)** and **Limiting Mag**.
2. In the Results card, click **Create Imaging Program**, name it, and click **Create**. Or, in the Imaging Log's **Programs** tab, click **New Program** and use **Catalog Pattern** with **M** and **110**.
3. Each month, set **Visibility** to the month, with **Catalog** still on Messier, to see which ones are in season.
4. Image each one as a project, and set it to **Completed** when it's done. The program's progress and the **Program Progress** report count it.

Make the Most of a Moonlit Night

A bright Moon doesn't have to mean a night off:

1. Open **Target Optimizer**.
2. Under **Source**, choose **To Do List**. The results update as soon as you choose.
3. Look at each card's **Moon** score. A high score means the Moon is down, faint, or far away during the target's window.
4. Prefer emission nebulae, especially with narrowband filters, and globular clusters. Save galaxies and reflection nebulae for dark nights.
5. In **Daily Visibility**, check your choice: the **Target-Moon Separation** card, and whether the background is dark while the white line is up.

Plan a Short Night or a Late Start

If you can't start until late, or must stop early:

1. In **Target Optimizer**, set **Start** to **Custom** and type the time you'll be ready. Targets that set before then drop down the list.

2. In **Sequence Planner**, set **Start** to **Custom** with the same time.
3. If you must stop before dawn, choose targets whose windows end by then; the Sequence Planner always plans until the last target sets or dawn.

Tune the Sequence Planner to Your Rig

The Sequence Planner's exposure counts are only as good as its session settings. To match them to your rig:

1. After a night, open **Log Analysis** and load its logs (see Log Analysis).
2. In the Combined Report's **Recommendations**, read the **Sequence Planning** table: **AF Duration**, **Guide Calibration Duration**, and **Flip Duration**.
3. In the Sequence Planner's **Session Settings**, set the same values.
4. If **Meridian Flip Verification** in **Data Quality** shows a large **Delta**, set **Flip Pause** and **Flip Offset** to match your ASIAir.

Do this after your first few nights, and again whenever you change your equipment. The sub gap and dither time take care of themselves.

Move to a New Computer

1. On the old computer, open the system menu (≡), click **Backup**, and save a **User Data** backup and a **Targets** backup.
2. Copy both files to the new computer.
3. Install Astryx there, or open the web version.
4. Click **Restore** and restore the targets backup, then the user data backup.

See Backup and Restore for details.

Chapter 17

Something Doesn’t Look Right

> **At a glance**
> Most surprises in Astryx have a simple cause: a setting that differs between views, a missing step, or no internet connection. Find what you’re seeing below.

Targets and Filters

The Visibility filter finds nothing, or Best Month is blank. Best Months haven’t been calculated for the location chosen in the sidebar, or the calculation was cancelled. Run **Calculate Best Months** from **Admin Tools** (see Admin Tools).

A target passes the Visibility filter, but the To Do List says it isn’t observable tonight. The Visibility filter looks at the whole month; the To Do List looks at tonight only. See the note in To Do List. If you’ve changed **Min Altitude** since Best Months were calculated, Target Details says so under **Criteria**; recalculate them (see Admin Tools).

A target I know is missing from the results. Check **Min Size (′)** and **Limiting Mag**: they start at your Settings, and can hide small or faint targets. An empty **Catalog** or **Type** list matches nothing. Try searching for the target by name instead.

A target has no Best Month at all. It never gets high enough from your location, or is high only in daylight or twilight. See Best Month Calculation.

Times and Dates

Times are an hour off. Check the location’s **Time Zone** in **Manage Observer Locations**. Astryx uses the location’s time zone and daylight saving, not your computer’s, so a location with the wrong time zone shows wrong times everywhere.

Times in the Target Optimizer or Sequence Planner don’t match Daily Visibility. Daily Visibility shows times in the location’s local time. The Target Optimizer and Sequence Planner show them in your computer’s time zone. They agree when your computer is in the location’s time zone.

Daily Visibility shows yesterday’s date. After midnight and before noon, it starts at the night you’re in, which began yesterday evening.

“No astronomical night at this location/date”. At high latitudes in summer, the Sun never gets 18° below the horizon, so there’s no astronomical darkness to plan in. Choose another date.

Planning

The Sequence Planner is empty, or says "No pinned targets". It plans only your Pinned Targets. Pin some in Target Selection, or with **Replace Pinned Targets** in the Target Optimizer.

My exposure lengths went back to 300 seconds. Changing the date, start, location, minimum altitude, or horizon setting, changing your Pinned Targets, or clicking **Reset & Optimize** builds a new plan with the default length. Set exposure lengths last.

A target was left out of the plan. With four or more targets, a target that can't get enough time above your minimum altitude is left out and listed under the Imaging Plan. Unpin it, or lower **Min Alt**.

The Target Optimizer ranks a target highly, but it's behind my trees. The Target Optimizer doesn't use your horizon profile. Check the target in Daily Visibility, which does.

The Daily Visibility button in the Target Optimizer shows the wrong night. The Optimizer's date goes back to today when you leave and return. Set the **Date** again before you click.

Online Features

The Viewfinder's sky image doesn't load. It needs an internet connection the first time you look at a target. Images you've seen recently are kept and work offline.

The forecast strips say "Cloud forecast unavailable for this date". The forecast covers today and the next four days only, and needs an internet connection.

Check for Target Updates says the file isn't available. It needs an internet connection.

Imaging Log

A session shows 0 exposures and 0.0h. Integration counts only **Used** exposures. Open the session and fill in **Used**.

A session won't save. It needs a **Date** that isn't in the future, a **Sub Length**, and at least one **Original Exposure**. If you record the session before imaging, enter the number you plan to take.

A finished target doesn't count in my program. Only projects with the status **Completed** count as imaged.

Log Analysis

There's no Combined Report. It needs both logs: the Session Log and the PHD2 Guide Log, from the same night.

There's no Meridian Flip Verification. It needs the night's location, which comes from an Imaging Log session with the same date. Add the session, then load the logs again.

The report doesn't know my telescope and sensor. As above: record the night in the Imaging Log first.

Data and Backups

My data is gone. On the web version, your data lives in the browser. Clearing the browser's data, using a private window, or using a different browser or computer shows an empty Astryx. Restore your latest backup (see Backup and Restore).

Automatic backups aren't happening. Check that **Automatic backups on change** is on in Settings. A backup is made only after you change your data, once **Minutes before backup** has passed. On the desktop app, backups go to the **Backup Folder** in Settings, or to your Downloads folder if it's empty.

A backup won't restore: "Version Mismatch". The backup was made by a version of Astryx with a different database version.

Still Stuck?

Describe what you see, and what you expected, in an issue at github.com/sparsileg/astryx/issues.

Appendix A

The Target Database

This appendix describes where Astryx's targets come from and what the database holds. The statistics are based on target database t1783038311, generated on July 3, 2026, which holds 14,009 targets.

Sources

The database was compiled from two main sources: the Saguaro Astronomy Club Deep Sky Database, and the OpenNGC database compiled by Mattia Verga, which also covers the IC catalog. Targets from other sources were merged in, and a supplementary file adds targets the main sources lack and extra details, such as common names, that make targets easier to find.

When the sources were merged, duplicate entries were combined so that no data was lost. An object that appears in several catalogs still has one entry per catalog, and each entry lists the others in its Other Info field. The Orion Nebula, for example, is both M 42 and NGC 1976. In all, 2,564 targets share an object with at least one other entry. Filter results show each such object once, under its most familiar catalog, so the same object doesn't fill the list twice. A name search still finds every entry (see Finding Targets).

Some targets have only one of the two sizes. Where a target had a maximum size but no minimum, or the reverse, the value it had was copied to the other. Those targets look circular, with the same minimum and maximum size.

Catalogs

Catalog	Targets
NGC	8,228
IC	4,746
Sharpless	313
Barnard	169
LDN (Lynds Dark Nebulae)	134
Caldwell	109
Messier	109
Abell	108
Extra	72

Catalog	Targets
Arp	19
Exotic	2

Types

Type	Targets
Galaxy	11,441
Open cluster	702
Emission nebula	533
Galaxy cluster	315
Dark nebula	304
Globular cluster	273
Planetary nebula	230
Cluster with nebulosity	79
Stars	57
Reflection nebula	53
Supernova remnant	22

Sizes

Every target has a size. The smallest is the planetary nebula IC 5117, at 0.02′ (about 1 arcsecond). The largest is the emission nebula Sh 2-109, at 1,080′ (18°).

Most targets are small, and most small targets are galaxies:

Size (arcminutes)	Targets	Share	Galaxies
Under 1	4,777	34.1%	93.5%
1 to 2	5,307	37.9%	91.2%
2 to 3	1,414	10.1%	87.1%
3 to 4	600	4.3%	72.5%
4 to 10	1,034	7.4%	36.8%
10 to 30	512	3.7%	16.0%
30 to 100	253	1.8%	0.8%
100 and over	112	0.8%	2.7%

82% of targets are under 3′. At the default Min Size of 4′, 1,911 targets remain. Above 10′, most targets are nebulae and star clusters, but 117 galaxies and galaxy clusters are larger than that, and some of them are superb imaging targets.

Magnitudes

4,862 targets, about a third, have no magnitude. They include 3,762 galaxies, 442 of the 533 emission nebulae, and all of the dark nebulae. The Limiting Mag filter always keeps targets that have no magnitude.

Using This

Unless you're hunting small galaxies, set a minimum size, and tick only the types you want to image. Otherwise galaxies fill the list. To come back to the same set of targets without filtering again, save it with **Create Imaging Program** (see Finding Targets).

Appendix B

Best Month Calculation

This appendix describes how Astryx works out each target's Best Month, Peak Altitude, and Observable months for a location. See Admin Tools for running the calculation.

The Year

The calculation covers the current calendar year, from January 1, at the location's latitude and longitude, in its time zone with its daylight saving. For each night it finds astronomical dusk and dawn (the Sun 18° below the horizon), and checks the target's altitude every 10 minutes in between.

Your horizon profile and the Moon aren't taken into account. For the Moon on a given night, use Yearly Observability or Daily Visibility.

Peak Altitude

The highest the target ever gets from the location, when it crosses the meridian:

```
peak altitude = 90° − |latitude − declination|
```

It's the same every night of the year.

Best Month

Each type of target has an altitude it must be above to count, and two weights:

Types	Altitude	Transit weight	Dark-hours weight
Single and double stars, asterisms	40°	0.75	0.25
Galaxies, dark and reflection nebulae	40°	0.70	0.30
Galaxy clusters, globular and open clusters	40°	0.65	0.35
Cluster with nebulosity, planetary nebulae	30°	0.60	0.40
Emission nebulae, supernova remnants	30°	0.55	0.45
Any other type	30°	0.60	0.40

Faint galaxies and nebulae need to be high, where there's less air to look through, so their transit weight is high. Bright emission nebulae and supernova remnants can be imaged lower down, and benefit more from long nights.

Every third night of the year gets two scores:

Transit score. How close to local midnight the target crosses the meridian:

```
transit score = 1 − (hours between transit and midnight) ÷ 12
```

A transit at midnight scores 1; one at 6 p.m. or 6 a.m. scores 0.5.

Dark-hours score. The hours the target spends above its type's altitude during astronomical darkness, divided by the most it gets on any night of the year.

They're combined:

```
score = transit score × transit weight + dark-hours score × dark-hours weight
```

The month of the night with the highest score is the Best Month.

For example, for a galaxy (0.70 and 0.30): a December night with the target crossing at 11 p.m. (transit score 0.92) and 8.5 dark hours out of a best of 9 (0.94) scores 0.92 × 0.70 + 0.94 × 0.30 = 0.93. A June night with the target crossing at midnight (1.0) but only 3.2 dark hours (0.36) scores 1.0 × 0.70 + 0.36 × 0.30 = 0.81. December wins: the long nights outweigh the slightly earlier transit.

Observable Months

The Observable months are the longest unbroken run of nights on which the target spends at least **2 hours in a row** above the **Min Altitude** in Settings during astronomical darkness. A run can wrap from December into January, for as in "Observable: Nov–Feb" in Yearly Observability.

The Best Month uses the type's own altitude, and the Observable months use your Min Altitude. That's why a target can have a Best Month but no Observable months, or the other way around.

When you change Min Altitude, the stored Observable months still use the old value until you recalculate. Astryx offers to recalculate the location in the sidebar when you save the change, and each other location when you choose it. Until then, Target Details shows the altitude they were calculated with under **Criteria**, and the new one beside it.

When an Astryx update changes how Best Months are worked out, Astryx recalculates every location that has them the next time it starts, the location in the sidebar first.

Targets with No Best Month

A target has no Best Month from a location when:

- **It never gets high enough.** Its peak altitude is below its type's altitude. From 39° N, a galaxy cluster at declination −27.5° peaks at 90° − 66.5° = 23.5°, well short of 40°. Most targets far south of a northern location, or far north of a southern one, are like this.
- **It's never high in darkness.** It reaches the altitude only in daylight or twilight, for example a target close to the Sun's path that is high only in the months of short summer nights at high latitudes.

The calculation's results count these as **Not meeting criteria**. They still appear in Target Selection, but never pass the Visibility filter.

Using the Best Month

- The Best Month is a guide, not a rule. A target is usually good for a month or two either side of it; the Observable months show how long.
- Your weather matters as much. The astronomically best month may be your cloudiest.
- Calculate again for a location you've moved by more than a few degrees of latitude.

Appendix C

Yearly Observability Details

This appendix describes how Yearly Observability works out the line and the background shading for each night. See Yearly Observability for how to read the graph.

The Nights

The graph covers 365 nights, starting on the first day of the current month. For each night, Astryx finds astronomical dusk and dawn (the Sun 18° below the horizon) at the location chosen in the sidebar, using the location's time zone and daylight saving time for that date.

The Line

The line is the target's highest altitude between dusk and dawn: its altitude as it crosses the meridian if that happens in the dark, otherwise the higher of its altitudes at dusk and at dawn. On nights with no astronomical darkness, as happens in summer at high latitudes, it's the target's altitude at local midnight instead.

The **Peak altitude** in the summary line is the highest point of the line over the whole year.

The Shading

Each night gets a score from 0 to 1, which sets the shade of gray: 0 is white and 1 is black. A night is white, whatever its score, when the line is below your Minimum Altitude.

The score has three parts.

Transit timing. How close to local midnight the target crosses the meridian:

```
transit score = 1 − (hours between transit and midnight) ÷ 12
```

A transit at midnight scores 1; one at 6 p.m. or 6 a.m. scores 0.5.

Dark hours. How many hours the target spends above a fixed altitude for its type during astronomical darkness, divided by the most hours it gets on any night of the year. The best night scores 1.

The two are combined with weights that also depend on the type:

```
base score = transit score × transit weight + dark-hours score × dark-hours weight
```

Types	Altitude	Transit weight	Dark-hours weight
Single and double stars, asterisms	40°	0.75	0.25
Galaxies, dark and reflection nebulae	40°	0.70	0.30
Galaxy clusters, globular and open clusters	40°	0.65	0.35
Cluster with nebulosity, planetary nebulae	30°	0.60	0.40
Emission nebulae, supernova remnants	30°	0.55	0.45
Any other type	30°	0.60	0.40

The Moon. The base score is multiplied by a Moon factor:

```
Moon factor = (1 − illumination) × (1 − e^(−separation ÷ 30°))
```

- *Illumination* is the Moon's illuminated fraction over the night, from 0 at new Moon to 1 at full.
- *Separation* is the closest the Moon comes to the target while the target is above its type's altitude, or over the whole night if it never gets that high.

The separation part is 0.63 at 30°, 0.86 at 60°, and 0.95 at 90°. A full Moon makes the factor 0, so full-Moon nights are white.

Finally, the score is raised to the power 0.7, which darkens the middle grays so the difference between fair and good nights is easier to see:

```
score = (base score × Moon factor)^0.7
```

The Full Moons

A gold circle marks each night when the Moon is more than 90% illuminated and brighter than on the nights before and after.

The Best Month and Observable Months

The Best Month and Observable months in the summary line aren't worked out by this view. They're the values stored for each target when the Best Months for a location were calculated, and they use the same type-specific altitudes and weights. See Best Month Calculation.

Accuracy

Target positions are the J2000 coordinates from the database, without correction for precession. For a graph that covers a year at a resolution of one day, the difference isn't visible.

Appendix D

Target Optimizer Scoring

This appendix describes how the Target Optimizer scores each target and each combination. See Target Optimizer for how to use it.

The Night

The night runs from the start you chose (astronomical dusk, or your Custom time) to astronomical dawn, at the location chosen in the sidebar. A target's **window** is the time it spends above both the Min Altitude from Settings and the location's horizon profile during the night.

A target is left out if it never gets above them, or if its window is shorter than one hour. If a target dips below them and comes back up, the window is one of the two parts; the dip is shown on the card but doesn't change the scores.

The Four Scores

Each score runs from 0 to 100.

Window. One hour scores 0 and eight hours scores 100:

```
window = (window hours − 1) ÷ 7 × 100, at most 100
```

Altitude. The peak is the highest altitude during the window, checked every 10 minutes:

```
altitude = (peak − Min Altitude) ÷ (90° − Min Altitude) × 100
```

Centering. How close the target's meridian crossing (transit) is to the middle of the night:

```
centering = (1 − |transit − middle of night| ÷ half the night's length) × 100
```

A target that doesn't cross the meridian during the night scores 0.

Moon. The window is checked every 15 minutes. Each check scores 1 if the Moon is below the horizon, and otherwise:

```
(1 − illumination) × separation ÷ 90°, with separation capped at 90°
```

Illumination is the Moon's illuminated fraction for the night, from 0 at new Moon to 1 at full; *separation* is the angle between the Moon and the target. The Moon score is the average of the checks, times 100.

The Composite Score

```
composite = 0.30 × window + 0.25 × altitude + 0.15 × centering + 0.30 × Moon
```

The results show the 37 targets with the highest composite scores.

Combinations

Combinations are built from the targets in the results. Each target in a combination is given **usable hours**: the hours of its window when no other target in the combination is up, plus an equal share of the hours it shares with them. Two targets up for the same two hours get one hour each of that time.

```
solo score     = composite × window hours
pair score     = (composite₁ × usable hours₁ + composite₂ × usable hours₂) ÷ 2
triplet score  = (sum of composite × usable hours over the three) ÷ 3
```

This is the **Combo score** on each combination. The five best solo targets, pairs, and triplets are shown.

Coverage is the time from the earliest window start to the latest window end in the combination, including any gaps between the windows.

Appendix E

Sequence Planner Rules

This appendix describes how the Sequence Planner builds a plan and counts exposures. See Sequence Planner for how to use it.

The Session

The night runs from astronomical dusk (the Sun 18° below the horizon), or your Custom start, to astronomical dawn. Within it, the session:

- starts when the first target in the plan is above your minimum altitude and horizon profile, if that's later than the start, and
- ends when the last target sets below them, if that's before dawn.

So the session's length depends on which targets come first and last, and the planner takes this into account when it chooses the order.

Overheads

At the start of each target: a guide calibration, then an autofocus (with **Autofocus** ticked).

Periodic autofocus: every **AF Interval** minutes after the last autofocus on the same target. The run waits for the exposure in progress to finish.

Meridian flip: a target that crosses the meridian during its turn flips at the transit plus the **Flip Offset**:

1. Imaging stops **Flip Pause** minutes before the flip.
2. The flip takes **Flip Duration** minutes.
3. An autofocus runs (with **Autofocus** ticked).
4. A guide calibration runs.

Counting Exposures

A target's turn is its share of the session. Its imaging time is the turn minus all the overheads in it:

```
exposures = imaging seconds ÷ (exposure length + time between exposures)
```

rounded down. The time between exposures is:

```
time between exposures = sub gap + dither time ÷ Frames per Dither
```

or just the sub gap with **Frames per Dither** at 0. The sub gap starts at 5 seconds and the dither time at 25 seconds. Once you analyze session logs in Log Analysis, each is the average of what your 20 most recent nights measured, counting only nights with enough clean samples.

The last exposure of the night is allowed to run past the end of the session rather than be cut short.

Ranking Plans

Plans are compared on **usable integration**: time spent on exposures while the target is above the minimum altitude and horizon. Time lost to overheads, or to a target that has set or sits behind the horizon, doesn't count. Each target's exposures are counted with its own exposure length.

Plans are ranked by, in order:

1. **Minimum integration**: the most targets with at least 120 minutes of usable integration. If not every target can get there, the plan that brings the weakest of the rest closest to 120 minutes wins.
2. **Least wasted time**: the most usable integration in total.
3. **Balance**: integration spread as evenly as possible across the targets.

Starting Order

Every plan starts from an equal split of the session, with the targets ordered by:

1. the time they set below your minimum altitude, soonest first, so targets about to be lost are imaged while they're still up;
2. then, for targets still up at the end of the session, the time they rise, with a target already up at the start counting as rising then;
3. then the time they cross the meridian, so each is imaged nearer its highest point;
4. then their names.

The order you pinned the targets in never matters.

Two or Three Targets

Order. Every order is tried: 2 for two targets, 6 for three. For each one, the session's start and end are worked out again, and the flip and handover steps below are applied. On a tie, the starting order wins.

Meridian flips. For each target that flips during its turn, three handovers to the next target are tried:

- **Exclude the flip**: hand over just before imaging would stop for the flip, so the flip never happens; the time saved goes to the next target.
- **Include the flip**: keep imaging through the flip and for a while after it, taking time from the next target.

- **Absorb the next target's flip**: when the next target would also flip, keep imaging the current one until after that flip time, so the next target starts on the right side of the meridian. One adjustment can save two flips.

Each is kept only if it ranks higher than the plan without it.

Handovers. Finally, each handover is moved in 1% steps to the position that ranks highest. A target keeps imaging until the next one has risen above the minimum altitude and horizon, or hands over early when it sets. The sweep repeats until no handover moves, since moving one can change the best place for another.

Four or More Targets

Trying every order stops being practical beyond three targets (24 orders for four, millions for ten), so larger plans are built another way:

1. **Leave out targets that can't be imaged.** A target that isn't above the minimum altitude and horizon for an equal share of the session, or for 120 minutes if that's less, is left out and listed as not planned. Leaving targets out raises everyone else's share, so the check repeats until no more drop out.
2. **Image the target that will be lost soonest.** Whenever a target finishes, the next is the one, of those already up, that sets soonest. The planner searches for the most usable imaging every target can get this way. A target that can't be fitted in at all is left out too.
3. **Avoid meridian flips.** If a target's turn would include its flip, it starts after the flip instead, and the target before it keeps imaging meanwhile. The flip is only paid for when waiting would lose the target, or for the night's first target, when waiting would leave the telescope idle.
4. **Hand out the time left over.** The handover sweep then moves the handovers to gain usable integration, without taking any target below what the weakest one got (or 120 minutes, if that's less).

Limits

- The Moon isn't taken into account.
- A target's turn isn't split around a horizon obstruction. The time behind the horizon is marked and doesn't count as usable integration, but the plan doesn't move another target into the gap.
- The planner can't talk to your equipment. The flip is modeled as a fixed block of time; the real timing depends on your capture software.
- With four or more targets, the order follows the set and rise times; other orders aren't tried, and time left over mostly goes to targets later in the night, since the evening targets are limited by when they set.

Appendix F

Combined Report Details

This appendix gives the rules behind the Combined Report in Log Analysis. See Log Analysis for how to read it.

Tiers

Each sub is compared with the night's median settled guide RMS: the middle value of all the subs' total RMS.

Tier	Rule
Unknown	No guide data covers the exposure, or the night has no median to compare with.
Reject	The exposure was cut short, or its RMS is more than 2 × the night's median.
Marginal	Its RMS is more than 1.2 × the median, it started before the dither settled, or a guide failure happened during it.
Clean	None of the above.

The rules are checked in that order, so a sub that is both cut short and unsettled is Reject.

Settled Guide RMS

The Verdict's guide RMS leaves out the time while each dither settles, since guiding is meant to be rough then. The Elevated guiding check uses these bands:

Settled RMS	Band
Under 0.95″	Excellent
0.95″ to 1.3″	Normal
1.3″ to 2″	Elevated
2″ to 4″	High
4″ or more	Critical

The Checks

Sixteen checks run on every night with light frames. Each finding has a severity: **Critical** needs attention, **Warning** is worth looking into, and **Info** is for the record.

Check	What it looks for
Guide-star swap	PHD2 jumping to a different star mid-session without saying so.
Cloud or transparency loss	A burst of guide failures: at least 3 in 5 minutes.
Unsettled start	Subs that started before the dither settled. Warning above 5% of the subs.
Truncated exposure	A sub that ended before its exposure time.
Manual intervention	An autorun stopped or an autofocus cancelled by hand, Plan Tonight paused, or a gap in the log.
Mount disconnect	The mount dropping its connection.
Frame cadence	Guide frames arriving at irregular intervals.
Elevated guiding	Settled RMS of 1.3″ or more: Info when elevated, Warning when high, Critical at 4″.
Axis-ratio inversion	Dec RMS higher than RA RMS, which often points to balance, backlash, or polar alignment.
Calibration outlier	PHD2 calibrations that disagree with each other, lost the star, or have a large orthogonality error.
Autofocus health	Autofocus failing more than usual, runs taking over 1.5 times the typical length, or star size growing through the night.
Plate-solve degradation	Plate solves that didn't center cleanly, or a falling star count.
Focus drift	The night's focus change per degree of temperature, when there are enough autofocus runs.
Dropped frames	The share of guide frames PHD2 dropped. Warning above 2%.
Lock position near edge	The guide star close to the edge of the guide camera's frame.
Mid-imaging guide recovery	Guiding lost and recovered in the middle of imaging.

Calibration outlier, Plate-solve degradation, and Mid-imaging guide recovery are new and not yet confirmed against many nights, so they report as Info whatever they find.

Internal Checks

Before the report is built, Astryx checks the two logs against each other: for example, that the number of dithers agrees, and that each sub's guide frames add up to its exposure time. A failed check appears in Data Quality with what it affects, and a mismatch that touches a Verdict figure is noted beside it.

Where the Limits Come From

The limits were set from the logs of one rig over nine months: 25 ASIAir logs and 19 PHD2 logs. They suit a typical small refractor on a strain-wave mount with a guide camera. A very different rig may trip some checks more, or less, often than it should.

Appendix G

The Command Line

The desktop app can also run a few tasks from a terminal. Each task does one thing Astryx already does, without opening its window, and then exits. You won't need them in normal use. They're for jobs that have no button in the app, or that are easier to repeat from a script.

The command line is in the desktop app only, not in Astryx on the web.

Running Astryx from a Terminal

Type `astryx` followed by the task's name. With no task, Astryx starts as usual.

- **Linux:** open a terminal and type, for example, `astryx --help`.
- **Windows:** open Command Prompt or PowerShell from the Start menu, then type the full path to Astryx, in quotes, followed by the task:
  - Command Prompt: `"%LOCALAPPDATA%\Astryx\astryx.exe" --help`
  - PowerShell: `& "$env:LOCALAPPDATA\Astryx\astryx.exe" --help`

  The Run box (Windows+R) isn't suitable: it has no window to show Astryx's messages in.

> **Note**
> On Windows, Astryx is a windowed app, so Command Prompt doesn't wait for it to finish. The prompt can come back before Astryx's message appears. Wait for the message, then press Enter for a fresh prompt.

Getting Help

- `astryx --help` or `astryx help` lists the tasks.
- `astryx help reset-learned` or `astryx reset-learned --help` describes one task.
- `astryx --version` shows which version of Astryx you have.

None of these opens Astryx. A mistyped task or option gets an error with a usage line instead.

Close Astryx First

A task writes to the same data as the open app. The open app would overwrite the change later, so a task refuses to run while Astryx is open:

```
Astryx is open; close it and try again
```

Close every Astryx window and run the task again.

Exit Codes

Every task finishes with a number, its exit code, which scripts can check:

Code	Meaning
0	The task succeeded.
1	The task failed, for example because Astryx was open. The message says why.
2	A command-line mistake, such as a mistyped task or option, or a PDF that isn't redirected to a file.

On Linux, `echo $?` shows the last exit code. On Windows, a batch file (`.cmd`) can check `%ERRORLEVEL%` after the task. Typed at the prompt it isn't reliable, because the prompt doesn't wait for Astryx.

Tasks

Task	What it does
`reset-learned`	Sets the learned Sub Gap and Dither Duration back to their defaults, and forgets the nights they were learned from. Use it after changing something that alters them, such as your camera, mount, dither size, or settle settings; loading your Session Logs again teaches Astryx your rig's values. It prints "Learned Sub Gap and Dither Duration reset to their defaults" when it's done.
`analyze-logs`	Analyzes one night's Session Log and PHD2 Guide Log, as loading them in Log Analysis does, and writes the Combined Report's PDF. See below.

Analyzing Logs

`analyze-logs` takes the night's two logs, in either order, and writes the same PDF as **Download PDF Report**. The PDF goes to the program's output, so send it to a file with `>`:

```
astryx analyze-logs Autorun_Log_2026-10-12_193023.txt PHD2_GuideLog_2026-10-12_195105.txt >
2026-10-12.pdf
```

Without `>`, Astryx refuses rather than fill the terminal with the PDF. To learn from the logs without keeping the report, send it to `/dev/null` on Linux or `NUL` on Windows.

As in Log Analysis, the Session Log adds its night to the learned Sub Gap and Dither Duration once. Astryx prints the result of each, in the words of the report's Behavior table:

```
Learned Sub Gap: 1.1s (was 1.2s; this log's 0.9s included, average of the last 20 nights)
Learned Dither Duration: 20.8s (was 20.9s; this log's 19.0s included, average of the last 20 nights)
```

Astryx refuses, with exit code 1 and nothing learned, when a file is missing or isn't a log, when both files are the same kind of log, or when the logs don't overlap in time. For the last, it prints when each log starts and ends. A night with only flats, darks, or bias frames has no report, so it also ends with exit code 1. When a task fails, the file after `>` is still created, empty; delete it.

Many Nights at Once

The learned figures are the average of your most recent nights by date, so the nights can be analyzed in any order.

On Linux, this script analyzes every Session Log in the current folder, with each guide log of the same date. Each report is named after its Session Log. A pair that doesn't overlap is refused, so only the right guide log produces a report:

```
#!/bin/sh
for session in Autorun_Log_*.txt; do
    day=$(echo "$session" | cut -d_ -f3)
    for guide in PHD2_GuideLog_"$day"_*.txt; do
        astryx analyze-logs "$session" "$guide" > report.tmp &&
            mv report.tmp "${session%.txt}.pdf"
    done
done
rm -f report.tmp
```

A guide log whose name has a different date from its Session Log's isn't found; analyze that night on its own.

On Windows, list the nights in a batch file (`.cmd`):

```
@echo off
set ASTRYX="%LOCALAPPDATA%\Astryx\astryx.exe"
%ASTRYX% analyze-logs Autorun_Log_2026-10-06_201425.txt PHD2_GuideLog_2026-10-06_200950.txt >
2026-10-06.pdf
%ASTRYX% analyze-logs Autorun_Log_2026-10-07_200436.txt PHD2_GuideLog_2026-10-07_193931.txt >
2026-10-07.pdf
```

A batch file waits for each night to finish before starting the next.

> **Note**
> On Windows, run `analyze-logs` from Command Prompt or a batch file. The PowerShell that comes with Windows (version 5.1) treats what comes through `>` as text and damages the PDF.

Starting the Learned Values Over

To make the learned Sub Gap and Dither Duration come from one night you trust, reset them and then analyze that night, discarding the report. On Linux:

```
#!/bin/bash
ASTRYX="/usr/bin/astryx"
LOGS="$HOME/logs"

"$ASTRYX" reset-learned
```

```
"$ASTRYX" analyze-logs "$LOGS/Autorun_Log_2025-10-23_202616.txt" "$LOGS/
PHD2_GuideLog_2025-10-23_201657.txt" > /dev/null
```

Add more `analyze-logs` lines to learn from more nights. On Windows, put the same two tasks in a batch file and send the report to `NUL`.

Appendix H

Accuracy

This appendix lists the approximations behind Astryx's calculations and how much they matter. In short: every time and position Astryx shows is far more precise than planning a night needs.

How Astryx Is Checked

Astryx's astronomy is tested against outside sources every time it changes:

- every astronomical dusk and dawn of 2026 at several test sites, against the U.S. Naval Observatory's tables,
- every new, first quarter, full, and last quarter Moon of 2026, against the same,
- a set of positions and times checked against other published values, and
- checks that views showing the same quantity, such as the Moon's illumination or dusk, agree with each other.

The Approximations

What	How it's done	How close
Target positions	The database's J2000 coordinates, used as they are, with no correction for precession or proper motion.	By 2026, up to about 0.4° from today's position: a minute or two in rise, set, and transit times, and a fraction of a degree in altitude.
The Sun	The Astronomical Almanac's low-precision formula.	About 0.01°; dusk and dawn to within a few seconds.
The Moon	A shortened version of the standard lunar theory.	About 0.3°; moonrise and moonset within a minute or two. Illumination to about 1%.
Moon phase names	Taken from the illumination.	Within about a day of the exact quarter.
Sidereal time	The standard J2000 formula.	Under 0.1 seconds for decades either side of 2000.

What	How it's done	How close
Altitudes	No allowance for atmospheric refraction, except at the horizon for sunrise, sunset, moonrise, and moonset.	Refraction lifts objects by less than 0.1° above 30°, where you image.
Rise and set times	Searched minute by minute.	Within one minute.
Hours above Min Altitude in darkness	Altitude every 10 minutes from dusk to dawn, with each rise or set found between samples.	Within a minute; a target up all night gets exactly the night's length.
Sky brightness	A model for comparing times and nights.	A ranking, not a measured sky brightness.

Time Zones

Each location uses its own time zone and daylight saving rules. Daylight saving is always taken as a one-hour shift; Lord Howe Island's half-hour shift is not modeled. Which side of a daylight saving change a night falls on is decided at local noon.

The Viewfinder's Coordinates

The center coordinates in the Viewfinder are J2000, like the target database. Most capture and plate-solving software accepts J2000.

Appendix I

Glossary

Altitude. How high something is above the horizon, in degrees: 0° on the horizon, 90° straight overhead.

Arcminute (′), arcsecond (″). Small angles: 60 arcminutes make a degree, and 60 arcseconds an arcminute. The full Moon is about 31′ across.

Astronomical dusk and dawn. When the Sun is 18° below the horizon, in the evening and the morning. Between them is astronomical darkness, when the sky is as dark as it gets.

Azimuth. The compass direction of something, in degrees: 0° north, 90° east, 180° south, 270° west.

Best Month. The month when a target is best placed for imaging from a location. See Best Month Calculation.

Bortle scale. A rating of sky darkness from 1, the darkest skies, to 9, an inner city.

Circumpolar. Never setting: always above the horizon, or above your minimum altitude, from your location.

Current Target. The target the Daily Visibility, Yearly Observability, and Viewfinder views show. See Introduction.

Declination (Dec). A target's position north or south of the celestial equator, in degrees, like latitude on Earth.

Dither. A small move of the mount between exposures, so that hot pixels and noise land in different places on each sub and stack out.

DSS. The Digitized Sky Survey, the photographic survey of the whole sky that the Viewfinder's sky images come from.

Field of view. How much sky your camera records through your telescope, in degrees.

Flip Offset, Flip Pause. How long after the target crosses the meridian the meridian flip starts, and how long before it imaging stops. See Sequence Planner.

Guide RMS. How far, on average, the guide star wandered from where the mount was trying to hold it, in arcseconds. Lower is better.

Horizon profile. The height of the trees, buildings, and hills around a location, in each direction. See Getting Started.

Integration time. The total exposure time of the subs you stack.

J2000. The standard reference for sky coordinates, fixed at the year 2000. The target database uses it.

Magnitude. A measure of brightness: the larger the number, the fainter the object.

Meridian. The line across the sky from north to south through the point overhead. A target is at its highest when it crosses it.

Meridian flip. On an equatorial mount, the turn of the telescope to the other side of the mount after a target crosses the meridian, so it can keep tracking.

Minimum altitude. The lowest altitude you're willing to image at. It's **Min Altitude** in Settings, and can be changed in several views.

Narrowband. Imaging through filters that pass only the light of a few gases, such as hydrogen-alpha. It cuts through moonlight and light pollution.

Pinned Targets. Your shortlist for the next night, listed in the sidebar. The Sequence Planner plans them.

Plate solving. Working out exactly where a telescope is pointing by matching the stars in an image against a catalog.

Right ascension (RA). A target's east–west position on the sky, in hours, minutes, and seconds, like longitude on Earth.

Seeing. How much the air blurs stars, usually in arcseconds. Good seeing is 2″ or less.

Sub. A single exposure, one of the many that are stacked into an image.

Sub gap. The time between the end of one sub and the start of the next, while the camera saves the image.

Transit. A target crossing the meridian, when it's at its highest.

Transparency. How clear the air is: how much haze, smoke, or thin cloud dims the stars.